MOKO

MKGW4 BLE Gateway

MQTT commands

Version1.2

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Contents

1. About This Manual

This manual describes the MQTT commands applied in MKGW4 gateway, will include the data structure and decoding examples for the actively reporting data, such as BLE scanning data, GNSS data. Also includes the inquiry and management commands which are used to read and configure gateway parameters from your cloud.

2. Command format

The MQTT protocol works in Hex format. It includes the setup commands, inquiry commands and actively reporting commands.

2.1 Setup gateway parameters (Write)

Send to gateway:

No.	Name	Description
1	Head	1 byte, 0XEF
2	Command flag	2 bytes, to indicate the command id and data writing. The command id starts from 0X1000.
3	MAC	6 bytes, gateway MAC address
4	Length	2 bytes, total data length
5	Data	n bytes, includes message TAG (1 byte) +message length (2 bytes) +message Value (m bytes)

Gateway reply:

No.	Name	Description
1	Head	1 byte, 0XEF
2	Command flag	2 bytes, starts from 0X1000.
3	MAC	6 bytes, gateway MAC address
4	Length	2 bytes. 0X0001
5	Data	0X01: command success 0X00: command false

Send to gateway: 0XEF + Command FLAG + MAC+ LENTH + DATA
Gateway reply: 0XEF+Command FLAG + MAC+ 0X0001 + 0X01/0X00
Note: Wrong command will not get a reply

2.2 Read gateway parameters (Read)

Send to gateway:

No.	Name	Description
1	Head	1 byte, 0XEF
2	Command flag	2 bytes, to indicate the command id and data reading The command id starts from 0X2000.
3	MAC	6 bytes, gateway MAC address
4	Length	2 bytes. 0X0000
5	Data	No data.

Gateway reply:

No.	Name	Description
1	Head	1 byte, 0XEF
2	Command flag	2 bytes, starts from 0X2000.
3	MAC	6 bytes, gateway MAC address
4	Length	2 bytes. Total data length
5	Data	n bytes, includes message TAG (1 byte) + message length(2 bytes) +message Value (m bytes)

Send to gateway: 0XEF + Command FLAG + MAC+ 0X0000
Gateway reply: 0XEF+Command FLAG + MAC+ LENTH + DATA
Note: Wrong command will not get a reply

2.3 Gateway data report (Report)

Gateway will actively report data to cloud in some situations. For example, reports BLE scanning data, GNSS data, gateway status, and so on.

No.	Name	Description
1	Head	1 byte, 0XEF
2	Command flag	2 bytes, to indicate the command id and data reporting. The command id starts from 0X3000.
3	MAC	6 bytes, gateway MAC address
4	Length	2 bytes. Total data length
5	Data	n bytes, includes message TAG (1 byte) + message length (2 bytes) +message Value (m bytes)

Gateways reports: 0XEF + Command FLAG + MAC+ LENGTH+DATA

3. Command list

3.1 System parameters

3.1.1 Reboot (0X1000)

To make the gateway run a software reboot.

Command flag	Massage tag	Massage tag description	Data description
0X1000	/	/	0X00 00

MQTT send: ef 10 00 dd 94 e2 66 56 11 00 00
Gateway reply: ef 10 00 dd 94 e2 66 56 11 00 01 01 //command success

3.1.2 Restore factory settings (0X1001)

To make the gateway restore to factory settings.

Command flag	Massage tag	Massage tag description	Data description
0X1001	/	/	0X00 00

MQTT send: ef 10 01 dd 94 e2 66 56 11 00 00
Gateway reply: ef 10 01 dd 94 e2 66 56 11 00 01 01 //command success

3.1.3 Power off (0X1002)

To power off the gateway.

Command flag	Massage tag	Massage tag description	Data description
0X1002	/	/	0X00 00

MQTT send: ef 10 02 dd 94 e2 66 56 11 00 00
Gateway reply: ef 10 02 dd 94 e2 66 56 11 00 01 01 //command success

3.1.4 Get device information(0X2003)

To inquire the basic device information from the gateway.

Command flag	Massage tag	Massage tag description	Data description
0X2003	0X00	Device name	string
	0X01	Product model	string
	0X02	Company name	string
	0X03	Hardware version	string
	0X04	Software version	string
	0X05	Firmware version	string
	0X06	IMEI	string
	0X07	ICCID	string

Send to gateway: ef 20 03 dd 94 e2 66 56 11 00 00
Gateway reply: ef 20 03 dd 94 e2 66 56 11 00 76 00 00 0a 4d 4b 47 57 34 2d 35 36 31 31 01 00 05 4d 4b 47 57 34 02 00 14 4d 4f 4b 4f 20 54 45 43 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e 03 00 04 56 31 2e 30 04 00 0d 6e 52 46 35 32 2d 53 44 4b 31 37 2e 30 05 00 07 56 31 2e 30 2e 32 54 06 00 0f 38 36 35 36 34 38 30 36 31 30 39 30 31 31 34 07 00 14 38 39 34 33 30 31 30 31 35 32 31 33 30 30 31 32 35 33 38 38

Description for the gateway reply:
Ef //head
20 03 //command flag
dd 94 e2 66 56 11 // gateway MAC
00 76 // total data length, 118 bytes
00 // device name flag
00 0a // device name length, 10 bytes
4d 4b 47 57 34 2d 35 36 31 31 // device name: MKGW4-5611
01 // product model flag
00 05 // product model length, 5bytes
4d 4b 47 57 34 //product model: MKGW4
02 //company name flag
00 14 //company name length, 20 bytes
4d 4f 4b 4f 20 54 45 43 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e //company name: MOKO TECHNOLOGY LTD.
03 //hardware version flag
00 04 // hardware version length, 4 bytes
56 31 2e 30 // hardware version: V1.0
04 //software version flag
00 0d // software version length, 13bytes
6e 52 46 35 32 2d 53 44 4b 31 37 2e 30 //software version: nRF52-SDK17.0
05 //firmware version flag
00 07 //firmware version length, 7 bytes
56 31 2e 30 2e 32 54 //firmware version: V1.0.2
06 //IMEI flag
00 0f // IMEI length, 15 bytes
38 36 35 36 34 38 30 36 31 30 39 30 31 31 34 //IMEI: 865648061090114
07 //ICCID flag
00 14 // iccid length,20 bytes
38 39 34 33 30 31 30 31 35 32 31 33 30 30 31 32 35 33 38 38 //ICCID: 89430101521300125388

3.1.5 Report device status (0X3004)

When the gateway successfully connects to cloud, it will immediately report this message once. If the gateway keeps a long connection with cloud, it will also report this message periodically.

Command flag	Massage tag	Massage tag description	Data description
0X3004	0X00	Timestamp	Hex. Unix timestamp, 4 bytes
	0X01	Network type	string. NB-IOT/CAT.M/GPRS
	0X02	CSQ	Hex, 1 byte, 0-31
	0X03	Battery voltage	Hex, 2 byte. Unit: mv
	0X04	3-axis sensor data	Hex, 6 bytes X-axis 2 bytes +Y-axis 2 bytes + 2 bytes Z-axis. unit: mg
	0X05	ACC status	1 byte. 0X00: on 0X01: off
	0X06	Imei	string

Gateway report:
ef 30 04 dd 94 e2 66 56 11 00 36 00 00 04 65 b8 b2 ec 01 00 04 47 50 52 53 02 00 01 1d 03 00 02 10 46 04 00 06 00 0a ff de ff fa 05 00 01 00 06 00 0f 38 36 35 36 34 38 30 36 31 30 39 30 31 31 34

Description for the gateway reporting data:
Ef // head
30 04 // command flag
dd 94e2 6656 11 // gateway MAC
00 36 // total data length, 54 Bytes
00 // timestamp value tag
0004 // data length, 4 Bytes
65b8 b2ec // timestamp value, 2024-01-30 16: 27: 24
01 // network type tag
00 04 // data length, 4 Bytes
47 5052 53 // network type, GPRS
02 // CSQ tag
0001 // data length, 1 Bytes
1d // CSQ, 29
03 // battery voltage tag
0002 // data length, 2 Bytes
1046 // battery voltage, 4166mV
04 // three axis sensor data tag
00 06 // data length, 6 Bytes

00 0a ff de ff fa // three axis sensor data, 10mg, 65502mg, 65530mg
05 // ACC status tag
0001 // data length, 1 Byte
00 // ACC status, off
06 // imei tag
000f // data length, 15 Byte
3836 3536 3438 3036 3130 3930 3131 34 // imei, 865648061090114

3.1.6 Firmware upgrade(0X1005)

To remotely upgrade the gateway firmware.

Command flag	Massage tag	Massage tag description	Data description
0X1005	0X00	Firmware file URL	String. 1-256characters

Send to gateway:
ef 10 05 dd 94 e2 66 56 11 00 3A 00 00 37 68 74 74 70 3A 2F 2F 34 37 2E 31 30 34 2E 31 37 32 2E 31 36 39 3A 38 30 38 30 2F 75 70 64 61 74 61 5F 66 6F 6C 64 2F 4D 4B 47 57 34 5F 56 31 2E 30 2E 32 2E 62 69 6E

gateway reply:
ef 10 05 dd 94 e2 66 56 11 00 01 01 //command success

Description for the sending command:
ef // head
10 05 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 3A // Total data length, 58 bytes
00 // URL flag
00 37 // URL length, 55 bytes
68 74 74 70 3A 2F 2F 34 37 2E 31 30 34 2E 31 37 32 2E 31 36 39 3A 38 30 38 30 2F 75 70 64 61 74 61 5F 66 6F 6C 64 2F 4D 4B 47 57 34 5F 56 31 2E 30 2E 32 2E 62 69 6E
// URL: http://47.104.172.169:8080/updata_fold/MKGW4_V1.0.2.bin

3.1.7 Firmware upgrade result(0X3006)

After the gateway firmware upgrade is complete, the gateway will actively report this message once.

Command flag	Massage tag	Massage tag description	Data description
0X3006	0X00	Firmware upgrade result	Hex, 1 byte -1: OTA in progress 0: OTA failed 1: OTA successful

Gateway report: Ef 30 06 dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
30 06 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 04 // total data length, 1 Byte
00 // firmware upgrade result tag
00 01 // data length, 1 Byte
01 // OTA successful

3.1.8 NTP setting(0X1007/0X2007)

To read and configure NTP settings for the gateway.

Command flag	Massage tag	Massage tag description	Data description
0X1007/ 0X2007	0X00	NTP switch	Hex, 1Byte. Range: 0/1, default: 1
	0X01	NTP server IP or domain name	string 0-64 characters
	0X02	NTP sync interval	Hex, 2Byte Range: 1-720, default: 24, unit: hour

Read command:

Send to gateway: ef 20 07 dd 94 e2 66 56 11 00 00
Device reply: ef 20 07 dd 94 e2 66 56 11 00 0c 00 00 01 01 01 00 00 02 00 02 00 18

Description for the gateway reply:
Ef // head
2007 // command flag
dd94e2665611 // gateway MAC
000c // total data length, 12 Byte
00 // NTP switch flag
0001 // data length, 1 Byte
01 // NTP switch, 01
01 // NTP server flag
0000 // data length, 0 Byte
02 // NTP sync interval flag
0002 // data length, 2 Byte
0018 // NTP sync interval: 24H

Set command:

Send to gateway: ef 10 07 dd 94 e2 66 56 11 00 0c 00 00 01 01 01 00 00 02 00 02 00 18
Device reply: ef 10 07 dd 94 e2 66 56 11 00 01 01 //command success

3.1.9 UTC time(0X1008/0X2008)

To read and configure UTC time for the gateway.

Command flag	Massage tag	Massage tag description	Data description
0X1008/ 0X2008	0X00	Unix timestamp	Hex, 4 bytes
	0X01	Time zone	Hex, 1-byte. Range: -24-28 (step by half time zone) Default to 0

Read command:

Send to gateway: ef 20 08 dd 94 e2 66 56 11 00 00
Device reply: ef 20 08 dd 94 e2 66 56 11 00 0b 00 00 04 65 b3 1b 7f 01 00 01 00

Description for the gateway reply:
Ef // head
20 08 // command flag
dd 94e2 6656 11 // gateway MAC
00 0b // total data length, 11 Byte
00 //timestamp tag
0004 //data length, 4 Byte
65 b3 1b7f //timestamp, 2024-01-26 10: 39: 59
01 //time zone tag
00 01 //data length, 1Byte
00 //time zone, 00

Set command:

Send to gateway: ef 10 08 dd 94 e2 66 56 11 00 0b 00 00 04 65 b3 1b 7f 01 00 01 00
Device reply: ef 10 08 dd 94 e2 66 56 11 00 01 01 //command success

3.1.10 Downlink communication timeout(0X1009/0X2009)

If there is no any data communication in the timeout after the gateway connects to a BLE device, the gateway will disconnect from the BLE device.

Command flag	Massage tag	Massage tag description	Data description
0X1009/ 0X2009	0X00	timeout	Hex, 1 byte Range: 0~60, unit: minute. if set to 0, the gateway will not disconnect from BLE. Default to 10

Read command:

App send: ef 20 09 fb c2 05 79 15 bb 00 00
Device reply: ef 20 09 fb c2 05 79 15 bb 00 04 00 00 01 0a

Description for the gateway reply:
Ef // head
2009 // command flag
fbc2057915bb // gateway MAC
0004 // total data length, 4 Byte
00 // timeout tag
0001 // data length, 1 Byte
0a // timeout 10 minutes

Set command:

App send: ef 10 09 fb c2 05 79 15 bb 00 04 00 00 01 0a
Device reply: ef 10 09 fb c2 05 79 15 bb 00 01 01 //command success

3.1.11 Indicator settings(0X100A/0X200A)

To read and configure the four LEDs on/off status.

Command flag	Massage tag	Massage tag description	Data description
0X100A/ 0X200A	0X00	led on/off	Hex, 1 byte Bit0: charge led on/off Bit1: switch led on/off Bit2: network led on/off Bit3: positioning led on/off Default to 0X0F

Read command:

Send to gateway: ef 20 0a fb c2 05 79 15 bb 00 00
Device reply: ef 20 0a fb c2 05 79 15 bb 00 04 00 00 01 0f

Description for the gateway reply:
Ef // head
200a // command flag
fbc2057915bb // gateway MAC
0004 // total data length, 4 Byte
00 // Indicator on/off switch tag
0001 // data length, 1 Byte
0f // charge led on, switch led on, network led on, positioning led on.

Set command:

Send to gateway: ef 10 0a fb c2 05 79 15 bb 00 04 00 00 01 0f
Device reply: ef 10 0a fb c2 05 79 15 bb 00 01 01 //command success

3.1.12 Certificate update or OTA status(0X200B)

To inquire if the gateway is upgrading firmware or downloading certificates.

Command flag	Massage tag	Massage tag description	Data description
0X200B	0X00	File download status	Hex, 1 byte 0: OTA or certificate update not performed 1: OTA or certificate is updating

Read command:

Send to gateway: ef 20 0b fb c2 05 79 15 bb 00 00
Device reply: ef 20 0b fb c2 05 79 15 bb 00 04 00 00 01 00

Description for the gateway reply:
200b // command flag
fbc2057915bb // gateway MAC
0004 // total data length, 4 Byte
00 // File download status flag
0001 // data length, 1 Byte
00 // OTA or certificate update not performed

3.1.13 Device status reporting setting(0X100C/0X200C)

To inquire and configure the device status reporting interval and reporting payload.

Command flag	Massage tag	Massage tag description	Data description
0X100C/ 0X200C	0X00	Device status report interval	Hex, 4 bytes 0/30-86400,unit: second 0 indicates reporting only once.
	0X01	Reporting payload selection	Hex, 1 byte Bit0: Battery voltage Bit1: 3-axis data Bit2: vehicle ACC status

Send to gateway: ef 20 0C fb c2 05 79 15 bb 00 00
Device reply: ef 20 0c fb c2 05 79 15 bb 00 0b 00 00 04 00 00 00 3c 01 00 01 07

Description for the gateway reply:
Ef // head
20 0c // command flag
fb c2 05 79 15 bb // gateway MAC
00 0b // total data length, 11 Byte
00 // Reporting interval tag
0004 // data length, 4 Byte

0000 003c //report interval: 60 seconds
01 // reporting payload selection tag
00 01 // data length, 1 Byte
07 // Battery voltage enable, 3-axis data enable, vehicle ACC status enable.

Set command:

Send to gateway: ef 10 0c fb c2 05 79 15 bb 00 0b 00 00 04 00 00 00 3c 01 00 01 07
Device reply: ef 10 0c fb c2 05 79 15 bb 00 01 01 //command success

3.1.14 Power disconnection alarm switch(0X100D/0X200D)

When the gateway detects the external power is disconnected, it will report an alarm message to cloud. This command is used to define the alarm reporting switch.

Command flag	Massage tag	Massage tag description	Data description
0x100D/ 0X200D	0X00	Alarm switch	Hex, 1 byte 0/1

Read command:

Send to gateway: ef 20 0d fb c2 05 79 15 bb 00 00
Device reply: ef 20 0d fb c2 05 79 15 bb 00 04 00 00 01 01 // command success

Description for the gateway reply:
Ef // head
20 0d // command flag
fb c2 05 79 15 bb // gateway MAC
00 04 // total data length, 4 Byte
00 // Power disconnect alarm switch tag
0001 // data length, 1 Byte
01 // Power disconnect alarm switch on

Set command:

Send to gateway: ef 10 0d fb c2 05 79 15 bb 00 00
Device reply: ef 10 0d fb c2 05 79 15 bb 00 01 01 // command success

3.1.15 Bluetooth connection password(0X100E/0X200E)

To read or configure the gateway Bluetooth password.

Command flag	Massage tag	Massage tag description	Data description
0X100E/ 0X200E	0X00	password	string 6-10 characters

Read command:

Send to gateway: ef 20 0e fb c2 05 79 15 bb 00 00
Device reply: ef 20 0e fb c2 05 79 15 bb 00 0b 00 00 08 4d 6f 6b 6f 34 33 32 31

Description for the gateway reply:
Ef // head
20 0e // command flag
fb c205 7915 bb // gateway MAC
00 0b // total data length, 11Byte
00 // password tag
0008 // data length, 8 Byte
4d6f 6b6f 3433 3231 // password Moko4321

Set command:

Send to gateway: ef 10 0e fb c2 05 79 15 bb 00 0b 00 00 08 4d 6f 6b 6f 34 33 32 31
Device reply: ef 10 0e fb c2 05 79 15 bb 00 01 01 // command success

3.1.16 Password verification(0X100F/0X200F)

To enable or disable the Bluetooth password verification.

Command flag	Massage tag	Massage tag description	Data description
0X100F/ 0X200F	0X00	Password verify switch	Hex, 1 byte Value: 0/1

Read command:

Send to gateway: ef 20 0f fb c2 05 79 15 bb 00 00
Device reply: ef 20 0f fb c2 05 79 15 bb 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 0f // command flag
fb c205 7915 bb // gateway MAC
00 04 // total data length, 4 Byte
00 // Password verification switch tag
0001 // data length, 1 Byte
01 // Password verification enabled

Setup command:

Send to gateway: ef 10 0f fb c2 05 79 15 bb 00 04 00 00 01 01 //enable password verification
Device reply: ef 10 0f fb c2 05 79 15 bb 00 01 01 //command success

3.1.17 Power disconnection alarm reporting(0X3010)

When the power disconnect alarm switch is on, the gateway detects the external power disconnects, it will report this message once.

Command flag	Massage tag	Massage tag description	Data description
0X3010	0X00	Timestamp	Hex, 4 bytes

Gateway reports: ef 30 10 fb c2 05 79 15 bb 00 07 00 00 04 65 b3 1b 7f

Description for the gateway reporting data:
Ef // head
3010 // command flag
fb c205 7915 bb // gateway MAC
00 07 // total data length, 7 Byte
00 // timestamp tag
0004 // data length, 4 Byte
65 b3 1b 7f // timestamp, 2024-01-26 10: 39: 59

3.1.18 Battery low voltage alarm(0X3011)

When the gateway detects the battery voltage is lower than the configured threshold, it will report an alarm message once.

Command flag	Massage tag	Massage tag description	Data description
0X3011	0X00	Timestamp	Hex, 4 bytes
	0X01	battery voltage	Hex, 2 bytes, unit: mV

Gateway reports: ef 30 11 fb c2 05 79 15 bb 00 0c 00 00 04 65 b3 1b 7f 01 00 02 0b b8

Description for the gateway reporting data:
Ef // head
3011 // command flag
fb c205 7915 bb // gateway MAC
00 0C // total data length, 12 Bytes
00 // timestamp tag
0004 // data length, 4 Byte
65 b3 1b 7f // timestamp, 2024-01-26 10: 39: 59
01 //battery voltage tag
00 02 // data length, 2 Byte
0b b8 // battery voltage: 3000mV

3.1.19 Battery low voltage parameters(0X1012/0X2012)

To inquire and configure the low battery voltage alarm switch and threshold.

Command flag	Massage tag	Massage tag description	Data description
0X 1012/ 0X2012	0X00	Battery low voltage alarm switch	Hex, 1 byte. Value: 0/1
	0X01	Low voltage threshold in percentage	Hex, 1 byte 0: 10% 1: 20% 2: 30% 3: 40% 4: 50%

Read command:

Send to gateway: ef 20 12 fb c2 05 79 15 bb 00 00
Device reply: ef 20 12 fb c2 05 79 15 bb 00 08 00 00 01 01 01 00 01 00

Description for the gateway reply:
Ef // head
20 12 // command flag
fb c205 7915 bb // gateway MAC
00 08 // total data length, 8 Byte
00 // switch tag
0001 // data length, 1 Byte
01 // switch on
01 // Low battery percentage
0001 // data length, 1 Byte
00 // Low battery percentage, 10%

Set command:

Send to gateway: ef 10 12 fb c2 05 79 15 bb 00 08 00 00 01 01 01 00 01 00
Device reply: ef 10 12 fb c2 05 79 15 bb 00 01 01 //command success

3.1.20 Inquire battery voltage(0X2013)

To inquire the gateway battery voltage.

Command flag	Massage tag	Massage tag description	Data description
0X2013	0X00	Battery voltage	Hex, 2 bytes unit mV

Read command:

Send to gateway: ef 20 13 fb c2 05 79 15 bb 00 00
Device reply: ef 20 13 fb c2 05 79 15 bb 00 05 00 00 02 0b b8

Description for the gateway reply:
Ef // head
20 13 // command flag
fb c2 05 79 15 bb // gateway MAC
00 05 // total data length,5 Byte
00 // device voltage tag
0002 // data length, 2 Byte
0b b8 // battery voltage: 3000 mV

3.1.21 Restore factory settings notification(0X3014)

When the gateway is reset by the physical button, this message will be reported once.

Command flag	Massage tag	Massage tag description	Data description
0X3014	/	/	0X00 00

Gateway report: ef 30 14 fb c2 05 79 15 bb 00 00

3.1.22 Power on when charging(0X1015/0X2015)

The gateway will be automatically turned on when it is charging. This command is used to set the automatic power on is enabled or disable.

Command flag	Massage tag	Massage tag description	Data description
0X1015/ 0X2015	0X00	Enable state	0/1

Read command:

Send to gateway: ef 20 15 fb c2 05 79 15 bb 00 00
Device reply: ef 20 15 fb c2 05 79 15 bb 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 15 // command flag
fb c2 05 79 15 bb // gateway MAC
00 04 // total data length, 4 Byte
00 //Automatic power on tag
0001 // data length, 1 Byte
01 // Automatic power on is enable.

Set command:

Send to gateway: ef 10 15 fb c2 05 79 15 bb 00 04 00 00 01 01
Device reply: ef 10 15 fb c2 05 79 15 bb 00 01 01 //command success

3.2 Network settings

3.2.1 Network configuration(0X1020/0X2020)

To remotely change the network settings. The new settings will take affect after a reboot.

Command flag	Massage tag	Massage tag description	Data description
0X1020/ 0X2020	0X00	Network priority	Hex, 1 byte 0. eMTC->NB-IOT->GSM 1. eMTC-> GSM -> NB-IOT 2. NB-IOT->GSM-> eMTC 3. NB-IOT-> eMTC-> GSM 4. GSM -> NB-IOT-> eMTC 5. GSM -> eMTC->NB-IOT 6. eMTC->NB-IOT 7. NB-IOT-> eMTC 8. GSM 9. NB-IOT 10. eMTC Default to 0
	0X01	APN	String. 0-100 characters
	0X02	APN username	String. 0-100 characters
	0X03	APN password	String. 0-100 characters

Read command:

Send to gateway: ef 20 20 fb c2 05 79 15 bb 00 00
Device reply: ef 20 20 fb c2 05 79 15 bb 00 0d 00 00 01 09 01 00 00 02 00 00 03 00 00

Description for the gateway reply:
20 20 // command flag
fb c205 7915 bb // gateway MAC
00 0d // total data length, 13 Byte
00 // Network priority tag
0001 // data length, 1 Byte
09 // Network priority: only NB-IOT
01 // APN tag
0000 // data length, 0 Byte
02 // APN user name tag
00 00 // data length, 0 Byte
03 // APN password tag

0000 // data length, 0 Byte

Set command:

Send to gateway: ef 10 20 fb c2 05 79 15 bb 00 0d 00 00 01 09 01 00 00 02 00 00 03 00 00
Device reply: ef 10 20 fb c2 05 79 15 bb 00 01 01 //command success

3.2.2 Connect network timeout(0X1021/0X2021)

If the gateway can not build a successful connection with cloud in the timeout, it will give up connection and enter sleep mode to save battery consumption. This command is used to configure the timeout.

Command flag	Massage tag	Massage tag description	Data description
0X1021 /0X2021	0X00	Connect timeout	Hex, 2 bytes 30-600, unit: second

Read command:

Send to gateway: ef 20 21 fb c2 05 79 15 bb 00 00
Device reply: ef 20 21 fb c2 05 79 15 bb 00 04 00 00 01 64

Description for the gateway reply:
Ef // head
20 21 // command flag
fb c205 7915 bb // gateway MAC
00 04 // total data length, 4 Bytes
00 // timeout tag
00 01 // data length, 1 Byte
64 // timeout: 100 seconds

Setup command:

Send to gateway: ef 10 21 fb c2 05 79 15 bb 00 04 00 00 01 64
Device reply: ef 10 21 fb c2 05 79 15 bb 00 01 01 //command success

3.3 MQTT settings

To remotely change the network settings. The new settings will take affect after a reboot.

3.3.1 MQTT parameters(0X1030/0X2030)

Command flag	Massage tag	Massage tag description	Data description
0X1030/ 0X2030	0X00	encryption mode	Hex, 1 byte 0: no encryption(TCP) 1: CA signed server

			certificate 2: CA certificate 3: self server certificates Default to 0
	0X01	Broker host	string 1~64 characters
	0X02	Broker port	Hex, 2 bytes 1~65535
	0X03	Client id	string 1~64 characters
	0X04	username	string 0-256 characters
	0X05	password	string 0-256 characters
	0X06	subscribe topic	string 1-128 characters
	0X07	publish topic	string 1-128 characters
	0X08	QOS	Hex, 1 byte 0-2
	0X09	clean session	Hex, 1 byte 0/1
	0X0A	keepalive	Hex, 1 byte 10-120, unit: second

Read command:

Send to gateway: ef 20 30 fb c2 05 79 15 bb 00 00
Device reply: ef 20 30 fb c2 05 79 15 bb 00 72 00 00 01 00 01 00 0c 34 37 2e 31 30 34 2e 38 31 2e 35 35 02 00 02 07 5b 03 00 0c 66 62 63 32 30 35 37 39 31 35 62 62 04 00 00 05 00 00 06 00 1b 2f 4d 4b 47 57 34 2f 66 62 63 32 30 35 37 39 31 35 62 62 2f 72 65 63 65 69 76 65 07 00 18 2f 4d 4b 47 57 34 2f 66 62 63 32 30 35 37 39 31 35 62 62 2f 73 65 6e 64 08 00 01 00 09 00 01 01 0a 00 01 3c

Description for the gateway reply:
Ef // head
20 30 // command flag
fb c205 7915 bb // gateway MAC
00 72 // total data length, 114 Byte
00 // encryption mode tag
0001 // data length, 1 Byte
00 // encryption mode, no encryption
01 // broker host tag
000c // data length, 12 Byte
3437 2e31 3034 2e38 312e 3535 // host: 47.104.81.55

02 // Broker port tag
00 02 // data length, 2 Byte
07 5b // broker port, 1883
03 // client_id tag
000c // data length, 12 Byte
6662 6332 3035 3739 3135 6262 // client id, fbc2057915bb
04 // user name tag
00 00 // data length, 0 Byte
05 // password tag
0000 // data length, 0 Byte
06 // subscribe topic tag
00 1b // data length, 11 Byte
2f 4d4b 4757 342f 6662 6332 3035 3739 3135 6262 2f72 6563 6569 7665 // subscribe topic: /MKGW4/fbc2057915bb/receive
07 // publish topic tag
00 18 // data length, 24 Byte
2f 4d4b 4757 342f 6662 6332 3035 3739 3135 6262 2f73 656e 64 // publish topic: /MKGW4/fbc2057915bb/send
08 // QOS tag
0001 // data length, 1 Byte
00 // QOS, 0
09 // clean session tag
0001 // data length, 1 Byte
01 // clean session, 1
0a // keepalive tag
0001 // data length, 1 Byte
3c // keepalive, 60 seconds

Set command:

Send to gateway: ef10 30 fb c2 05 79 15 bb 00 72 00 00 01 00 01 00 0c 34 37 2e 31 30 34 2e 38 31 2e 35 35 02 00 02 07 5b 03 00 0c 66 62 63 32 30 35 37 39 31 35 62 62 04 00 00 05 00 00 06 00 1b 2f 4d 4b 47 57 34 2f 66 62 63 32 30 35 37 39 31 35 62 62 2f 72 65 63 65 69 76 65 07 00 18 2f 4d 4b 47 57 34 2f 66 62 63 32 30 35 37 39 31 35 62 62 2f 73 65 6e 64 08 00 01 00 09 00 01 01 0a 00 01 3c
Device reply: ef10 30 fb c2 05 79 15 bb 00 01 01 //command success

3.3.2 SSL certificate update(0X1031)

When change the MQTT settings, if it requires SSL certificates, the gateway will obtained the files from a HTTP file server. This command is send the certificates file URL to the gateway.

Command flag	Massage tag	Massage tag description	Data description
0X1031	0X00	CA file url	String. each URL is 0-256 characters The three files can not be all empty
	0X01	Client certificate url	
	0X02	Client key url	

Send to gateway: ef 10 31 fb c2 05 79 15 bb 00 36 00 00 2d 68 74 74 70 3A 2F 2F 34 37 2E 31 30 34 2E 31 37 32 2E 31 36 39 3A 38 30 38 30 2F 75 70 64 61 74 61 5F 66 6F 6C 64 2F 63 61 2E 63 65 72 74 01 00 00 02 00 00
Device reply: ef 10 31 fb c2 05 79 15 bb 00 01 01 //command success

Description for the gateway reply:
Ef // head
10 31 // command flag
fb c205 7915 bb // gateway MAC
00 36 // total data length, 54 Bytes
00 // CA file URL tag
00 2d // data length, 45 Bytes
68 74 74 70 3A 2F 2F 34 37 2E 31 30 34 2E 31 37 32 2E 31 36 39 3A 38 30 38 30 2F 75 70 64 61 74 61 5F 66 6F 6C 64 2F 63 61 2E 63 65 72
74//http://47.104.172.169:8080/updata_fold/ca.cert
01 // client certificate file URL tag
00 00 // data length, 0 byte
02 // private key file URL tag
00 00 // data length, 0 byte

3.3.3 MQTT certificate download result(0X3032)

When the gateway finishes to download the certificates, it will send this message.

Command flag	Massage tag	Massage tag description	Data description
0X3032	0X00	Update results	Hex, 1 byte -1: In progress 0: Failed 1: Success

Gateway reports:
Ef 30 32 fb c2 05 79 15 bb 04 00 00 01 01 //certificates downloaded successfully

3.4 Scan and report parameters

3.4.1 Scan and report parameter (0X1040/0X2040)

To set the parameters related to BLE scan and data report mode.

The gateway supports multiple modes, when the gateway is working in Real time scan & immediate report, and it detects the external power is disconnected, it will automatically switch to Real time scan &periodic report to reduce battery consumption. The automatic switch function can be set enabled or disabled.

Command flag	Massage tag	Massage tag description	Data description
0X1040 /0X2040	0X00	Scan& report mode	Hex, 1 byte 0: Scan turned off 1: Real time scan & immediate report 2: Real time scan &periodic report 3: Periodic scan & immediate reporting 4: Periodic scan & periodic report Default to 2
	0X01	Mode automatic switch	Hex, 1 byte 0/1

Read command:

Send to gateway: ef 20 40 dd 94 E2 66 56 11 00 00
Device reply: ef 20 40 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 00

Description for the gateway reply:
Ef // head
20 40 // command flag
dd 94e2 6656 11 // gateway MAC
00 08 // total data length, 8 Byte
00 // Scan and report mode tag
0001 // data length, 1 Byte
01 // Real time scan& immediate report
01 //Automatic mode switch tag
0001 // data length, 1 Byte
00 //Automatic mode switching is off

Set command:

Send to gateway: ef 10 40 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 00
Device reply: ef 10 40 dd 94 e2 66 56 11 00 01 01 //command success

3.4.2 Real time scan &periodic report parameters (0X1041/0X2041)

In the real time scan & periodic report mode, the gateway will report BLE scanning data as the configured interval.

Command flag	Massage tag	Massage tag description	Data description
0X1041/ 0X2041	0X00	Reporting interval	Hex, 4 bytes 600-86400, unit: seconds

Read command:

Send to gateway: ef 20 41 dd 94 E2 66 56 11 00 00
Device reply: ef 20 41 dd 94 e2 66 56 11 00 07 00 00 04 00 00 02 58

Description for the gateway reply:
Ef // head
20 41 // command flag
dd 94e2 6656 11 // gateway MAC
00 07 // total data length, 7 Byte
00 //reporting interval tag
0004 // data length, 4 Byte
0000 0258 //reporting interval, 600 seconds

Set command:

Send to gateway: ef 10 41 dd 94 e2 66 56 11 00 07 00 00 04 00 00 02 58
Device reply: ef 10 41 dd 94 e2 66 56 11 00 01 01 //command success

3.4.3 Periodic scan&immediate report parameters(0X1042/0X2042)

To set the BLE scan duration and interval for the periodic scan and immediate report mode.

Command flag	Massage tag	Massage tag description	Data description
0X1042/ 0X2042	0X00	Scan duration	Hex, 2 bytes 3-3600, unit: second
	0X01	Scan interval	Hex, 4 bytes 600-86400, unit: second

Read command:

Send to gateway: ef 20 42 dd 94 E2 66 56 11 00 00
Device reply: ef 20 42 dd 94 e2 66 56 11 00 0c 00 00 02 00 3c 01 00 04 00 00 02 58

Description for the gateway reply:
Ef // head
20 42 // command flag
dd 94e2 6656 11 // gateway MAC
00 0c // total data length, 12 Byte
00 // Scan duration tag
0002 // data length, 2 Byte

003c // Scan duration, 60s
01 // scan interval tag
00 04 // data length, 4 Byte
00 0002 58 // scan interval, 600s
Set command:

Send to gateway: ef 10 42 dd 94 e2 66 56 11 00 0c 00 00 02 00 3c 01 00 04 00 00 02 58
Device reply: ef 10 42 dd 94 e2 66 56 11 00 01 01 //command success

3.4.4Periodic scan &periodic report parameters(0X1043/0X2043)

To set the BLE scan duration and interval, and report interval for the periodic scan and periodic report mode.

Command flag	Massage tag	Massage tag description	Data description
0X1043/ 0X2043	0X00	Scan duration	Hex, 2 bytes 3-3600, seconds
	0X01	Scan interval	Hex, 4 bytes 600-86400, seconds
	0X02	Report interval	Hex, 4 bytes 600-86400, seconds
	0X03	Data retention priority	Hex, 1 byte 0: next period 1: current period

Read command:

Send to gateway: ef 20 43 dd 94 E2 66 56 11 00 00
Device reply: ef 20 43 dd 94 e2 66 56 11 00 17 00 00 02 00 3c 01 00 04 00 00 02 58 02 00 04 00 00 02 58 03 00 01 00

Description for the gateway reply:
Ef // head
20 43 // command flag
dd 94e2 6656 11 // gateway MAC
00 17 // total data length, 23 Byte
00 // scan duration tag
0002 // data length, 2 Byte
003c // scan duration, 60s
01 //scan interval tag
00 04 // data length, 4 Byte
00 0002 58 //scan interval, 600s
02 // report interval tag
0004 // data length, 4 Byte
0000 0258 // report interval, 600s
03 // Bluetooth data retention priority tag

00 01 // data length, 1 Byte
00 // Ble data retention priority:next period

Set command:

Send to gateway: ef 10 43 dd 94 e2 66 56 11 00 17 00 00 02 00 3c 01 00 04 00 00 02 58 02 00 04 00 00 02 58 03 00 01 00
Device reply: ef 10 43 dd 94 e2 66 56 11 00 01 01 //command success

3.5 BLE scan filter settings

3.5.1 Filtration logic(0X1050/0X2050)

To set the process logic of MAC address, ADV name and Raw data filtration.

Command flag	Massage tag	Massage tag description	Data description
0X1050/ X02050	0X00	Filters processing logic	Hex, 1 byte 0: None 1: MAC 2: ADV NAME 3: RAW DATA 4: ADV NAME & RAW DATA 5: MAC & ADV NAME & RAW DATA 6: ADV NAME | RAW DATA 7: ADV NAME & MAC Default: 4

Send to gateway: ef 20 50 dd 94 e2 66 56 11 00 00
Device reply: ef 20 50 dd 94 e2 66 56 11 00 04 00 00 01 01 //command success

Ef // head
20 50 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter logic tag
0001 // data length, 1 Byte
01 // filter logic, process by MAC filter

Set command:

Send to gateway: ef 10 50 dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 50 dd 94 e2 66 56 11 00 01 01 //command success

3.5.2 RSSI filtration(0X1051/0X2051)

Filter BLE data by Bluetooth signal strength RSSI.

Command flag	Massage tag	Massage tag description	Data description
0X1051/ 0X2051	0X00	RSSI	1 byte, number signed value, range -127~0. Default: -127

Read command:

Send to gateway: ef 20 51 dd 94 e2 66 56 11 00 00
Device reply: ef 20 51 dd 94 e2 66 56 11 00 04 00 00 01 81

Description for the gateway reply:
Ef // head
20 51 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter RSSI tag
0001 // data length, 1 Byte
81 // filter RSSI, -127 dBm

Set command:

Send to gateway: ef 10 51 dd 94 e2 66 56 11 00 04 00 00 01 81
Device reply: ef10 51 dd 94 e2 66 56 11 00 01 01 //command success

3.5.3 PHY filtration(0X1052/0X2052)

Filter BLE data by advertising physical layer.

Command flag	Massage tag	Massage tag description	Data description
0X1052/ 0X2052	0X00	PHY	number, 1 byte 0: 1M PHY(V4.2) 1: 1M PHY(V5.0) 2: 1M PHY(V4.2) & 1M PHY(V5.0) 3: Coded PHY(V5.0) Default to 0

Read command:

Send to gateway: ef 20 52 dd 94 e2 66 56 11 00 00
Device reply: ef 20 52 dd 94 e2 66 56 11 00 04 00 00 01 00

Description for gateway reply:
Ef // head

20 52 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter PHY tag
0001 // data length, 1 Byte
00 // filter PHY, 1M PHY (BLE V4.2)

Set command:

Send to gateway: ef 10 52 dd 94 e2 66 56 11 00 04 00 00 01 00
Device reply: ef 10 52 dd 94 e2 66 56 11 00 01 01 //command success

3.5.4 MAC filtration(0X1053/0X2053)

Filter BLE data by mac address.

Command flag	Massage tag	Massage tag description	Data description
0X1053 /0X2053	0X00	Precision filtering switch	number, 1 byte 0/1
	0X01	Reverse filter switch	number, 1 byte 0/1
	0X02	Filter MAC address, Each expression includes tag, length and MAC value, up to 10 groups mac can be set.	Hex, each mac is 1-6 bytes

Read command:

Send to gateway: ef 20 53 dd 94 e2 66 56 11 00 00
Device reply: ef 20 53 dd 94 e2 66 56 11 00 12 00 00 01 00 01 00 01 00 02 00 02 dd 6f 02 00 02 c4 1e

Description for gateway reply:
Ef // head
20 53 // command flag
dd 94e2 6656 11 // gateway MAC
00 12// total data length, 18 Bytes
00 // precision filtering switch tag
00 01// data length, 1 byte
00 // precision filtering is disabled
00 // reverse filter switch, off
01 // reverse filter tag
0001 // data length, 1 Byte
00 // reverse filtering is disabled
02 // mac address tag

00 02 // first mac length
dd6f // first mac
02 // mac address tag
00 02// second mac length
c4 1e // second mac address

Set command:

Send to gateway: ef10 53 dd 94 e2 66 56 11 00 0d 00 00 01 00 01 00 01 00 02 00 02 c4 1e
Device reply: ef10 53 dd 94 e2 66 56 11 00 01 01 //command success

3.5.5 ADV name filtration(0X1054/0X2054)

Filter BLE data by advertising name.

Command flag	Massage tag	Massage tag description	Data description
0X1054/ 0X2054	0X00	Precision filtering switch	number, 1 byte 0/1
	0X01	Reverse filter switch	number, 1 byte 0/1
	0X02	Filter adv name Each expression includes tag, length and name value, up to 10 groups name can be set.	string each advertising name is 1-20 characters

Read command:

Send to gateway: ef 20 54 dd 94 E2 66 56 11 00 00
Device reply: ef 20 54 dd 94 e2 66 56 11 00 1d 00 00 01 00 01 00 01 00 02 00 09 4d 4b 20 42 75 74 74 6f 6e 02 00 06 4d 4b 20 50 49 52

Ef // head
20 54 // command flag
dd 94e2 6656 11 // gateway MAC
00 1d // total data length, 29 Byte
00 // Precision filtering switch tag
0001 // data length, 1 Byte
00 // Precision filtering switch, off
01 // Reverse filter switch tag
0001 // data length, 1 Byte
00 // Reverse filter switch off
02 // adv name tag
00 09// first adv name length, 9 byte
4d4b 2042 7574 746f 6e // first adv name: MK Button

02 // adv name tag
00 06 // first adv name length, 6 byte
4d4b 2050 495 // first adv name: MK PIR

Set command:

Send to gateway: ef 10 54 dd 94 e2 66 56 11 00 08 00 00 01 00 01 00 01 00 02 00 00
Device reply: ef 10 54 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6 Raw data filtration(0X1055/0X2055)

Filter BLE data by the advertising raw data type.

Command flag	Massage tag	Massage tag description	Data description
0X1055/ 0X2055	0X00	Filter switch for BLE data types	Hex, 2 bytes Bit0: ibeacon Bit1: eddystone-uid Bit2: eddystone-url Bit3: eddystone-tlm Bit4: bxp-devinfo Bit5: bxp-acc Bit6: bxp-th Bit7: bxp-button Bit8: bxp-tag Bit9: pir Bit10: tof Bit11: other default 0X0FFF

Read command:

Send to gateway: ef 20 55 dd 94 E2 66 56 11 00 00
Device reply: ef 20 55 dd 94 e2 66 56 11 00 05 00 00 02 0f ff

Description for gateway reporting data:
Ef // head
20 55 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // filter switch tag
0002 // data length, 2 Byte
0fff // all types on

Set command:

Send to gateway: ef 10 55 dd 94 e2 66 56 11 00 05 00 00 02 0f ff
Device reply: ef 10 55 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.1 iBeacon filtration parameters (0X1056/0X2056)

Inquire and set filter parameters for iBeacon data.

Command flag	Massage tag	Massage tag description	Data description
0X1056/ 0X2056	0X00	Filter switch	number, 1 byte 0/1
	0X01	min major	number, 2 bytes Max major>=min major
	0X02	max major	
	0X03	min minor	number, 2 bytes Max minor>=min minor
	0X04	max minor	
	0X05	UUID	hex, 0-16 bytes

Read command:

App send: ef 20 56 dd 94 E 26 65 611 00 00
Device reply: ef 20 56 dd 94 e2 66 56 11 00 1b 00 00 01 01 01 00 02 00 00 02 00 02 ff ff 03 00 02 00 00 04 00 02 ff ff 05 00 00

Description for the gateway reply:
Ef // head
20 56 // command flag
dd 94e2 6656 11 // gateway MAC
00 1b // total data length, 27 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on
01 // min major tag
0002 // data length, 2 Byte
0000 // min major, 0
02 // max major tag
00 02 // data length, 2 Byte
ff ff // max major, 65535
03 // min minor tag
0002 // data length, 2 Byte
0000 // min minor, 0
04 // max minor tag
00 02// data length, 2 Byte
ff ff // max minor, 65535
05 // uuid tag
0000 // data length, 0 Byte

Set command:

App send: ef 10 56 dd 94 e2 66 56 11 00 1b 00 00 01 01 01 00 02 00 00 02 00 02 ff ff 03 00 02 00 00 04 00 02 ff ff 05 00 00
Device reply: ef 10 56 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.2 Eddystone-uid filtration parameter(0X1057/0X2057)

Inquire and set filter parameters for Eddystone-UID data.

Command flag	Massage tag	Massage tag description	Data description
0X1057/ 0X2057	0X00	Filter switch	number, 1 byte 0/1
	0X01	namespace	hex, 0-10 byte
	0X02	instance	Hex, 0-6 byte

Read command:

App send: ef 20 57 dd 94 E2 66 56 11 00 00
Device reply: ef 20 57 dd 94 e2 66 56 11 00 0a 00 00 01 01 01 00 00 02 00 00

Description for the gateway reply:
Ef // head
20 57 // command flag
dd 94e2 6656 11 // gateway MAC
00 0a // total data length, 1 Byte
00 // Filter switch tag
0001 // data length, 1 Byte
01 // Filter switch, on
01 // namespace tag
0000 // data length, 0 Byte
02 // instance tag
00 00 // data length, 0 Byte

Set command:

App send: ef 10 57 dd 94 e2 66 56 11 00 0a 00 00 01 01 01 00 00 02 00 00
Device reply: ef 10 57 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.3 Eddystone-URL filtration parameters(0X1058/0X2058)

Inquire and set filter parameters for Eddystone-URL data.

Command flag	Massage tag	Massage tag description	Data description
0X1058/ 0x2058	0X00	Filter switch	number, 1 byte 0/1
	0X01	url	string, 0-37 characters

Read command:

App send: ef 20 58 dd 94 E2 66 56 11 00 00
Device reply: ef 20 58 dd 94 e2 66 56 11 00 07 00 00 01 01 01 00 00

Description for the gateway reply:
Ef // head
20 58 // command flag
dd 94e2 6656 11 // gateway MAC
00 07 // total data length, 1 Byte
00 // Filter switch tag
0001 // data length, 1 Byte
01 // Filter switch, on
01 // url tag
0000 // data length, 0 Byte

Set command:

App send: ef 10 58 dd 94 e2 66 56 11 00 07 00 00 01 01 01 00 00
Device reply: ef 10 58 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.4 Eddystone-TLM filtration parameters(0X1059/0X2059)

Inquire and set filter parameters for Eddystone-TLM data.

Command flag	Massage tag	Massage tag description	Data description
0X1059/ 0X2059	0X00	Filter switch	number, 1 byte 0/1
	0X01	TLM version	number, 1 byte 0: version 0 1: version 1 2: all versions

Read command:

App send: ef 20 59 dd 94 E2 66 56 11 00 00
Device reply: ef 20 59 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 02

Description for the gateway reply:
Ef // head
20 59 // command flag
dd 94e2 6656 11 // gateway MAC
00 08 // total data length, 8 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on
01 // TLM version tag
0001 // data length, 1 Byte
02 // all TLM versions

Set command:

App send: ef10 59 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 02
Device reply: ef 10 59 dd 94 e2 66 56 11 00 01 01//command success

3.5.6.5 BXP-Device info filtration parameters(0X105A/0X205A)

Inquire and set filter parameters for MOKO defined BXP-device info data.

Command flag	Massage tag	Massage tag description	Data description
0X105A/ 0X205A	0X00	Filter switch	number, 1 byte 0/1

Read command:

App send: ef 20 5a dd 94 e2 66 56 11 00 00
Device reply: ef 20 5a dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 5a // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on

Set command:

App send: ef 10 5a dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 5a dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.6 BXP- ACC filtration parameters(0X105B/0X205B)

Inquire and set filter parameters for MOKO defined BXP-ACC data.

Command flag	Massage tag	Massage tag description	Data description
0X105B/ 0X205B	0X00	Filter switch	Value type: number, 1 byte 0/1

Read command:

App send: ef 20 5b dd 94 e2 66 56 11 00 00
Device reply: ef 20 5b dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:

Ef // head
20 5b // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 1 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on

Set command:

App send: ef 10 5b dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 5b dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.7 BXP-TH filtration parameters(0X105C/0X205C)

Inquire and set filter parameters for MOKO defined BXP-T&H data.

Command flag	Massage tag	Massage tag description	Data description
0X105C/ 0X205C	0X00	Filter switch	number, 1 byte 0/1

Read command:

App send: ef 20 5c dd 94 e2 66 56 11 00 00
Device reply: ef 20 5c dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 5c // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on

Set command:

App send: ef 10 5c dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 5c dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.8 BXP- Button filtration parameters(0X105D/0X205D)

Inquire and set filter parameters for MOKO defined BXP-Button data.

Command flag	Massage tag	Massage tag description	Data description
0X105D/ 0X205D	0X00	Filter switch	number, 1 byte 0/1
	0X01	Filter rules	hex, 1 byte Bit0: single press alarm advertisement

			Bit1: double press alarm advertisement Bit2: long press alarm advertisement Bit3: Abnormal inactivity alarm advertisement

Read command:

App send: ef 20 5d dd 94 e2 66 56 11 00 00
Device reply: ef 20 5d dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 0f

Description for the gateway reply:
Ef // head
20 5d // command flag
dd 94e2 6656 11 // gateway MAC
00 08 // total data length, 8 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on
01 // filter rules tag
0001 // data length, 1 Byte
0f // single press, double press ,long press and abnormal inactivity advertisement are all on

Set command:

App send: ef 10 5d dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 0f
Device reply: ef 10 5d dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.9 BXP- Tag filtration parameters(0X105E/0X205E)

Inquire and set filter parameters for MOKO defined BXP-Tag data.

Command flag	Massage tag	Massage tag description	Data description
0X105E/ 0X205E	0X00	Filter switch	number, 1 byte 0/1
	0X01	Precision filter switch	number, 1 byte 0/1
	0X02	Reverse filter switch	number, 1 byte 0/1
	0X03	Filter TAG ID (each expression includes the tag, length and value, up to 10 TAG IDs can be set)	hex each tag id is 1-6 bytes

Read command:

App send: ef 20 5e dd 94 e2 66 56 11 00 00
Device reply: ef 20 5e dd 94 e2 66 56 11 00 0c 00 00 01 01 01 00 01 00 02 00 01 00

Description for the gateway reply:
Ef // head
20 5e // command flag
dd 94e2 6656 11 // gateway MAC
00 0c // total data length, 12 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on
01 // Precision filtering switch tag
0001 // data length, 1 Byte
00 // Precision filtering switch, off
02 // Reverse filter switch tag
0001 // data length, 1 Byte
00 // Reverse filter switch, off

Set command:

App send: ef 10 5e dd 94 e2 66 56 11 00 0c 00 00 01 01 01 00 01 00 02 00 01 00
Device reply: ef 10 5e dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.10 MK PIR filtration parameters(0X105F/0X205F)

Inquire and set filter parameters for MOKO defined MK PIR data.

Command flag	Massage tag	Massage tag description	Data description
0X105F/ 0X205F	0X00	Filter switch	number, 1 byte 0/1
	0X01	Delay response status	number, 1 byte 0: low delay 1: medium delay 2: high delay 3: all types Default to 3
	0X02	Door status	number, 1 byte 0: close 1: open 2: all types Default to 2
	0X03	Sensor sensitivity	number, 1 byte 0: low sensitivity 1: medium sensitivity 2: high sensitivity 3: all types Default to 3

	0X04	Sensor detection status	number, 1 byte 0: no effective motion detected 1: effective motion detected 2: all types Default to 2
	0X05	min major	number, 2 bytes Max major>=min major
	0X06	max major	
	0X07	min minor	number, 2 bytes Max minor>=min minor
	0X08	max minor	

Read command:

App send: ef 20 5f dd 94 e2 66 56 11 00 00
Device reply: ef 20 5f dd 94 e2 66 56 11 00 28 00 00 01 01 01 00 01 03 02 00 01 02 03 00 01 03 04 00 01 02 05 00 02 00 00 06 00 02 ff ff 07 00 02 00 00 08 00 02 ff ff

Description for the gateway reply:
Ef // head
20 5f // command flag
dd 94e2 6656 11 // gateway MAC
00 28 // total data length, 40 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on
01 // delay response status tag
0001 // data length, 1 Byte
03 // delay response status, all types
02 // door status tag
0001 // data length, 1 Byte
02 // door status, all types
03 // sensor sensitivity tag
0001 // data length, 1 Byte
03 // sensor sensitivity, all types
04 // sensor detection_status tag
0001 // data length, 1 Byte
02 // sensor detection status, all types
05 // min major tag
0002 // data length, 1 Byte
0000 // min major, 0
06 // max major tag
00 02 // data length, 2 Byte
ff ff // max major, 65535

07 // min minor tag
0002 // data length, 2 Byte
0000 // min minor,0
08 //max minor tag
00 02 // data length, 2 Byte
ff ff //max minor 65535

Set command:

App send: ef 10 5f dd 94 e2 66 56 11 00 28 00 00 01 01 01 00 01 03 02 00 01 02 03 00 01 03 04 00 01 02 05 00 02 00 00 06 00 02 ff ff 07 00 02 00 00 08 00 02 ff ff
Device reply: ef 10 5f dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.11 MK TOF filtration parameters(0X1060/0X2060)

Inquire and set filter parameters for MOKO defined MK TOF data.

Command flag	Massage tag	Massage tag description	Data description
0X1060/ 0X2060	0X00	Filter switch	number, 1 byte 0/1
	0X01	Filter MFG code (each expression includes the tag, length and code value, up to 10 MFG codes can be set)	hex, Each MFG code is 2 bytes

Read command:

App send: ef 20 60 dd 94 E2 66 56 11 00 00
Device reply: ef 20 60 dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 60 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // filter switch tag
0001 // data length, 1 Byte
01 // filter switch, on

Set command:

App send: ef 10 60 dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 60 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.12 Other type filtration parameters(0X1061/0X2061)

Inquire and set filter parameters for other types BLE advertising data.

Command flag	Massage tag	Massage tag description	Data description
0X1061/ 0X2061	0X00	Filter switch	number, 1 byte 0/1
	0X01	Processing logic of the below filters	number, 1 byte 0: A 1: A&B 2: A | B 3: A&B&C 4: (A&B) | C 5: A | B | C Default to 5
	0X02	Filter rules, up to 3 groups of filter rules can be set. Each expression includes tag, length and value. The value includes: data type + start byte + end byte + raw data value	Data type: hex, 1 byte, 0X00~0XFF Start byte: number, 1byte End byte: number, 1 byte Raw data value: hex, 1-29 byte

Read command:

App send: ef 20 61 dd 94 E2 66 56 11 00 00
Device reply: ef 20 61 dd 94 e2 66 56 11 00 19 00 00 01 01 01 00 01 05 02 00 05 FF 01 02 4C 00 02 00 06 16 01 03 59 00 08

Description for the gateway reply:
Ef // head
20 61 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 19 // total data length, 25 Byte
00 // filter switch tag
00 01 // data length, 1 Byte
01 // filter switch, on
01 // filter logic tag
00 01 // data length, 1 Byte
05 // processing logic, A | B | C
02 // filter rule tag
00 05 // data length of the first rule
FF 01 02 4C 00 // data type: 0xFF, start byte: 01, end byte: 02, raw data value: 4C 00
02 // filter rule tag
00 06 // data length of second rule

16 01 03 59 00 08 // data type 0x16, start byte: 01, end byte: 03, raw data value: 59 00 08

Set command:

App send: ef 10 61 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 05
Device reply: ef 10 61 dd 94 e2 66 56 11 00 01 01 //command success

3.5.6.13 Duplicate data filtering rule(0X1062/0X2062)

The gateway can filter out duplicate packets to reduce transmission data flow. This command is to set the rule to judge duplicate packets.

Command flag	Massage tag	Massage tag description	Data description
0X1062/ 0X2062	0X00	rule to judge duplicate packets	number, 1 byte 0: filter disabled; 1: MAC; 2: MAC+Data type; 3: MAC+RAW Data; Default: 0

Read command:

Send to gateway: ef 20 62 dd 94 E2 66 56 11 00 00
Device reply: ef 20 62 dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 62 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // rule tag
0001 // data length, 1 Byte
01 // rule, judge by MAC address

Set command:

Send to gateway: ef 10 62 dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 62 dd 94 e2 66 56 11 00 01 01 //command success

3.6 Bluetooth advertising parameters

3.6.1 Advertising parameters (0X1070/0X2070)

To inquire and set the gateway advertising parameters.

Command flag	Massage tag	Massage tag description	Data description
0X1070/ 0X2070	0X00	Response packet switch	hex, 1 byte 0/1

	0X01	Advertising name	string, 1-10 characters
	0X02	Advertising interval	number, 1 byte 1-100, unit:100ms
	0X03	TX power	number, 1 byte Value: -40, -20, -16, -12, -8, -4, 0, 2, 3, 4, 5, 6, 7, 8. unit: dBm
	0X04	Advertising timeout	number, 1 byte 0-60, unit:min

Read command:

App send: ef 20 70 dd 94 E2 66 56 11 00 00
Device reply: ef 20 70 dd 94 e2 66 56 11 00 1d 00 00 01 01 01 00 0a 4d 4b 47 57 34 2d 35 36 31 31 02 00 01 0a 03 00 01 00 04 00 01 00

Description for the gateway reply:
Ef // head
20 70 // command flag
dd 94e2 6656 11 // gateway MAC
00 1d // total data length, 29 Byte
00 // response packet switch tag
0001 // data length, 1 Byte
01 // response packet switch on
01 // advertising name tag
000a // data length, 10 Byte
4d4b 4757 342d 3536 3131 // advertising name, MKGW4-5611
02 //advertising interval tag
00 01 // data length, 1 Byte
0a // advertising interval, 1000ms
03 // TX power tag
00 01 // data length, 1 Byte
00 // Tx power, 0 dBm
04 // advertising timeout tag
00 01 // data length, 1 Byte
00 // advertising timeout, 0

Set command:

App send: ef 10 70 dd 94 e2 66 56 11 00 1d 00 00 01 01 01 00 0a 4d 4b 47 57 34 2d 35 36 31 31 02 00 01 0a 03 00 01 00 04 00 01 00
Device reply: ef 10 70 dd 94 e2 66 56 11 00 01 01 //command success

3.6.2 iBeacon advertising parameters(0X1071/0X2071)

To inquire and set the gateway iBeacon advertising parameters.

Command flag	Massage tag	Massage tag description	Data description
0X1071/ 0X2071	0X00	major	number, 2 bytes
	0X01	minor	number, 2 bytes
	0X02	uuid	Hex, 16 bytes
	0X03	rssi@1m	number, 1 byte

Read command:

App send: ef 20 71 dd 94 E2 66 56 11 00 00
Device reply: ef 20 71 dd 94 e2 66 56 11 00 21 00 00 02 00 00 01 00 02 00 00 02 00 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03 00 01 c5

Description for the gateway reply:
Ef // head
20 71 // command flag
dd 94e2 6656 11 // gateway MAC
00 21 // total data length, 33 Byte
00 // major tag
0002 // data length, 2 Byte
0000 // major, 0
01 //minor tag
00 02 // data length, 2 Byte
0001//minor, 1
02 // uuid tag
0010 // data length, 16 Byte
0000 0000 0000 0000 0000 0000 0000 0000 // uuid
03 // rssi@1m tag
00 01 // data length, 1 Byte
c5 // rssi@1m, -59 dBm

Set command:

App send: ef 10 71 dd 94 e2 66 56 11 00 21 00 00 02 00 00 01 00 02 00 00 02 00 10 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03 00 01 c5
Device reply: ef 10 71 dd 94 e2 66 56 11 00 01 01 //command success

3.7 Positioning parameters

3.7.1 Positioning Mode(0X1080/0X2080)

The gateway supports multiple positioning fix modes. This command is used to inquire and configure the mode.

Command flag	Massage tag	Massage tag description	Data description
0X1080/ 0X2080	0X00	fix mode	Hex, 1 byte 0: Turn off position fix 1: Periodic fix 2: Motion fix

Read command:

Send to gateway: ef 20 80 dd 94 E2 66 56 11 00 00
Device reply: ef 20 80 dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
20 80 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 1 Byte
00 // fix mode tag
0001 // data length, 1 Byte
01 // fix mode, periodic positioning

Set command:

Send to gateway: ef 10 80 dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 80 dd 94 e2 66 56 11 00 01 01 //command success

3.7.1 Fix interval for periodic fix(0X1081/0X2081)

To set the fix interval for periodic fix mode.

Command flag	Massage tag	Massage tag description	Data description
0X1081/ 0X2081	0X00	fix interval	number, 4 bytes 60-86400. unit: second

Read command:

Send to gateway: ef 20 81 dd 94 E2 66 56 11 00 00
Device reply: ef 20 81 dd 94 e2 66 56 11 00 07 00 00 04 00 00 0e 10

Description for the gateway reply:
Ef // head
20 81 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 07 // total data length, 7 Byte
00 // interval tag
00 04 // data length, 4 Byte
00 00 0e 10 // interval, 3600 seconds

Set command:

Send to gateway: ef 10 81 dd 94 e2 66 56 11 00 07 00 00 04 00 00 0e 10
Device reply: ef 10 81 dd 94 e2 66 56 11 00 01 01 //command success

3.7.2 3-Axis parameters(0X1082/0X2082)

To set the parameters for the accelerometer sensor.

Command flag	Massage tag	Massage tag description	Data description
0X1082/ 0X2082	0X00	wake-up threshold	number, 1 byte 1-20, unit:16mg
	0X01	wake-up duration	number, 1 byte 1-10, unit:10ms
	0X02	motion threshold	number, 1 byte 10-250, unit: 2mg
	0X03	motion duration	number, 1 byte 1-15, unit: 5ms

Read command:

Send to gateway: ef 20 82 dd 94 E2 66 56 11 00 00
Device reply: ef 20 82 dd 94 e2 66 56 11 00 10 00 00 01 03 01 00 01 02 02 00 01 10 03 00 01 08

Description for the gateway reply:
Ef // head
20 82 // command flag
dd 94e2 6656 11 // gateway MAC
00 10 // total data length, 16 Byte
00 // wake-up threshold tag
0001 // data length, 1 Byte
03 // wake-up threshold, 3*16mg=48mg
01 // wake-up duration tag
0001 // data length, 1 Byte
02 //wake-up duration, 2*10ms=20ms
02 //motion detection threshold tag
0001 // data length, 1 Byte
10 //motion threshold, 16*2mg=32mg
03 //motion detection duration tag
0001 // data length, 1 Byte
08 //motion detection duration, 8*5ms=40ms

Set command:

Send to gateway: ef 10 82 dd 94 e2 66 56 11 00 10 00 00 01 03 01 00 01 02 02 00 01 10 03 00 01 08
Device reply: ef 10 82 dd 94 e2 66 56 11 00 01 01 //command success

3.7.3 Motion start fix parameters(0X1083/0X2083)

To enable or disable the GPS fix when it detect motion starts.

Command flag	Massage tag	Massage tag description	Data description
0X1083/ 0X2083	0X00	Fix when motion starts	number, 1 byte 0: disable 1: enable

Read command:

Send to gateway: ef 20 83 dd 94 E2 66 56 11 00 00
Device reply: ef 20 83 dd 94 e2 66 56 11 00 04 00 00 01 01

Description for the gateway reply:
Ef // head
20 83 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // positioning switch tag
00 01 // data length, 1 Byte
01 // fix when motion start is enabled

Set command:

Send to gateway: ef 10 83 dd 94 e2 66 56 11 00 04 00 00 01 01
Device reply: ef 10 83 dd 94 e2 66 56 11 00 01 01 //command success

3.7.4 Fix parameters in journey(0X1084/0X2084)

To enable or disable the GPS fix and interval when the gateway is moving in the journey.

Command flag	Massage tag	Massage tag description	Data description
0X1084/ 0X2084	0X00	Fix in journey	number, 1 byte 0: disable 1: enable
	0X01	Fix interval	number, 4 bytes 10-86400, unit: seconds

Read command:

Send to gateway: ef 20 84 dd 94 E2 66 56 11 00 00
Device reply: ef 20 84 dd 94 e2 66 56 11 00 0b 00 00 01 01 01 00 04 00 00 01 2c

Description for the gateway reply:
Ef // head
20 84 // command flag

dd 94e2 6656 11 // gateway MAC
00 0b // total data length, 10 Byte
00 // fix in journey tag
0001 // data length, 1 Byte
01 // fix in journey is enabled
01 // fix interval tag
0004 // data length, 4 Byte
0000 012c // fix interval, 300 seconds

Set command:

Send to gateway: ef 10 84 dd 94 e2 66 56 11 00 0b 00 00 01 01 01 00 04 00 00 01 2c
Device reply: ef 10 84 dd 94 e2 66 56 11 00 01 01 //command success

3.7.5 Motion stops fix parameters(0X1085/0X2085)

To enable or disable the GPS fix when motion stops and set the motion stop duration.

Command flag	Massage tag	Massage tag description	Data description
0X1085/ 0X2085	0X00	fix when motion stops	1 byte, number. 0: disable 1: enable
	0X01	motion stop duration	1 byte, number. range: 3-180, unit 10 seconds default 12, 120 seconds

Read command:

Send to gateway: ef 20 85 dd 94 E2 66 56 11 00 00
Device reply: ef 20 85 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 0c

Description for the gateway reply:
Ef // head
20 85 // command flag
dd 94e2 6656 11 // gateway MAC
00 08 // total data length, 8 Byte
00 // motion stop fix tag
0001 // data length, 1 Byte
01 // motion stop fix is enabled
01 // motion stop duration tag
0001 // data length, 1 Byte
0c // motion stop duration, 120 seconds

Set command:

Send to gateway: ef 10 85 dd 94 e2 66 56 11 00 08 00 00 01 01 01 00 01 0c
Device reply: ef 10 85 dd 94 e2 66 56 11 00 01 01 //command success

3.7.6 Fix parameters in stationary(0X1086/0X2086)

To enable or disable the GPS fix and interval when the gateway is not moving, and stays in stationary.

Command flag	Massage tag	Massage tag description	Data description
0X1086/ 0X2086	0X00	fix in stationary	1 byte, number. 0: disable 1: enable
	0X01	fix interval	1 byte,number Range: 1-1440, unit: minute, default 60

Read command:

Send to gateway: ef 20 86 dd 94 E2 66 56 11 00 00
Device reply: ef 20 86 dd 94 e2 66 56 11 00 09 00 00 01 01 01 00 02 00 3c

Description for the gateway reply:
Ef // head
20 86 // command flag
dd 94e2 6656 11 // gateway MAC
00 09 // total data length, 9 Byte
00 // fix in stationary tag
0001 // data length, 1 Byte
01 // fix in stationary is enabled
01 // fix interval tag
0002 // data length, 2 Byte
003c // fix interval, 60 minutes

Set command:

Send to gateway: ef 10 86 dd 94 e2 66 56 11 00 09 00 00 01 01 01 00 02 00 3c
Device reply: ef 10 86 dd 94 e2 66 56 11 00 01 01 //command success

3.7.7 GPS parameters(0X1087/0X2087)

To set the GPS parameters.

Command flag	Massage tag	Massage tag description	Data description
0X1087/ 0X2087	0X00	fix timeout	number, 2 bytes Range: 60-600, unit: second
	0X01	PDOP	number, 1 byte Range: 25-100, unit: 0.1

Read command:

Send to gateway: ef 20 87 dd 94 E2 66 56 11 00 00
Device reply: ef 20 87dd 94 e2 66 56 11 00 09 00 00 02 00 b4 01 00 01 64

Description for the gateway reply:
Ef // head
20 87 // command flag
dd 94e2 6656 11 // gateway MAC
00 09 // total data length, 9 Byte
00 // fix timeout tag
0002 // data length, 2 Byte
00b4 // fix timeout, 180s
01 // PDOP tag
00 01 // data length, 1 Byte
64 // PDOP, 100*0.1=10

Set command:

Send to gateway: ef 10 87 dd 94 e2 66 56 11 00 09 00 00 02 00 b4 01 00 01 64
Device reply: ef 10 87 dd 94 e2 66 56 11 00 0101 //command success

3.7.8 Request current location (0X1088)

To inquire the gateway current location, the gateway will do a GPS fix and report the location immediately. The location information will be reported in the 0X3089 packet.

Command flag	Massage tag	Massage tag description	Data description
0X1088	/	/	/

Send to gateway: ef 10 88 dd 94 E2 66 56 11 00 00
Device reply: ef 10 88 dd 94 e2 66 6 11 00 01 01 //command success

3.7.9 GNSS data reporting(0X3089)

When the gateway gets a GNSS fix, it will actively report the fix result and location to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3089	0X00	Timestamp	4 bytes, Unix timestamp
	0X01	fix mode	1 byte, number 0: Periodic fix 1: motion fix 2: Request fix
	0X02	fix result	1 byte, number 0: GPS fix successful

			1: LBS fix successful 2: Fix failed cause by the request fix interruption. 3: Fix failed caused by GPS serial port is occupied 4: Fix failed caused by GPS timeout 5: Fix failed caused by GPS fix timeout 6: Fix failed caused by PDOP limited 7: LBS fix failed
	0X03	latitude and longitude	8 bytes, number. It includes 4 bytes longitude and 4 bytes latitude with an accuracy of 0.0000001. Signed value, negative values indicate west longitude and south latitude
	0X04	LBS information	Hex, 6 bytes The first 2 bytes are LAC/TAC The last 4 bytes are CI

Gateway report:
Ef 30 89 fb c2 05 79 15 bb 00 1a 00 00 04 65 c0 b3 42 01 00 01 00 02 00 01 00 03 00 08 44 3a 9f ec 12 27 89 66

Description for gateway reporting data:
Ef // head
3089 // command flag
fb c2 05 79 15 bb // gateway MAC
00 1a // total data length, 26 Bytes
00 // timestamp tag
0004 // data length, 4 Byte
65c0 b342 // 2024-02-05 18:06:58
01 // fix mode tag
00 01 // data length, 1 Byte
00 // fix mode: period fix
02 // fix result tag
00 01 // data length, 1 byte
00 // GPS fix successful
03 // latitude and longitude tag

0008 // data length, 8 byte
44 3a9f ec12 2789 66 // 114.4692716; 30.4580966

3.8 Upload payload settings

3.8.1 iBeacon packet(0X1090/0X2090)

To configure the uploading payload for the iBeacon packet.

Command flag	Massage tag	Massage tag description	Data description
0X1090/ 0X2090	0X00	iBeacon packet upload	1 byte, hex to binary Bit0: RSSI Bit1: Timestamp Bit2: uuid Bit3: major Bit4: minor Bit5: rssi@1m Bit6: Advertising raw data Bit7: Response raw data

Read command:

App send: ef 20 90 dd 94 E2 66 56 11 00 00
Device reply: ef 20 90 dd 94 e2 66 56 11 00 04 00 00 01 3f

Description for the gateway reply:
Ef // head
20 90 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // iBeacon payload tag
0001 // data length, 1 Byte
3f // 0011 1111 to indicate message in bit 0 to bit 5 are uploaded.

Set command:

App send: ef 10 90 dd 94 e2 66 56 11 00 04 00 00 01 3f
Device reply: ef 10 90 dd 94 e2 66 56 11 00 01 01 //command success

3.8.2 Eddystone-UID packet(0X1091/0X2091)

To configure the uploading payload for the eddystone-uid packet.

Command flag	Massage tag	Massage tag description	Data description
0X1091/ 0X2091	0X00	eddystone-uid packet upload	1 byte, hex to binary Bit0: RSSI Bit1: timestamp

			Bit2: rssi@0m Bit3: namespace Bit4: instance Bit5: advertising raw data Bit6: response raw data

Read command:

App send: ef 20 91 dd 94 E2 66 56 11 00 00
Device reply: ef 20 91 dd 94 e2 66 56 11 00 04 00 00 01 1f

Description for the gateway reply:
Ef // head
20 91 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // eddystone-uid upload tag
0001 // data length, 1 Byte
1f //0001 1111 to indicate message in bit 0 to bit 4 are uploaded.

Set command:

App send: ef 10 91 dd 94 e2 66 56 11 00 04 00 00 01 1f
Device reply: ef 10 91 dd 94 e2 66 56 11 00 01 01 //command success

3.8.3 Eddystone-URL packet(0X1092/0X2092)

To configure the uploading payload for the Eddystone-url packet.

Command flag	Massage tag	Massage tag description	Data description
0X1092/ 0X2092	0X00	Eddystone-url packet upload	1 byte, hex to binary Bit0: RSSI Bit1: timestamp Bit2: rssi@0m Bit3: URL Bit4: advertising raw data Bit5: response raw data

Read command:

App send: ef 20 92 dd 94 E2 66 56 11 00 00
Device reply: ef 20 92 dd 94 e2 66 56 11 00 04 00 00 01 0f

Description for the gateway reply:
Ef // head
20 92 // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // Eddystone-url upload tag
0001 // data length, 1 Byte
0f // 0000 1111 to indicate message in bit 0 to bit 3 are uploaded.

Set command:

App send: ef 10 92 dd 94 e2 66 56 11 00 04 00 00 01 0f
Device reply: ef 10 92 dd 94 e2 66 56 11 00 01 01 //command success

3.8.4 Eddystone-TLM packet (0X1093/0X2093)

To configure the uploading payload for the Eddystone-tlm packet.

Command flag	Massage tag	Massage tag description	Data description
0X1093/ 0X2093	0X00	Eddystone-tlm packet upload	2 bytes,hex to binary Bit0: RSSI Bit1: timestamp Bit2: TLM version Bit3: battery voltage Bit4: chip temperature Bit5: ADV CNT Bit6: SEC CNT Bit7: advertising raw data Bit8: response raw data

Read command:

App send: ef 20 93 dd 94 E2 66 56 11 00 00
Device reply: ef 20 93 dd 94 e2 66 56 11 00 05 00 00 02 00 7f

Description for the gateway reply:
Ef // head
20 93 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // Eddystone-tlm upload tag
0002 // data length, 2 Byte
007f // 0111 1111 to indicate message in bit 0 to bit 6 are uploaded.

Set command:

App send: ef 10 93 dd 94 e2 66 56 11 00 05 00 00 02 00 7f
Device reply: ef 10 93 dd 94 e2 66 56 11 00 01 01 //command success

3.8.5 BXP-Devinfo packet(0X1094/0X2094)

To configure the uploading payload for the MOKO defined BXP-device info packet.

Command flag	Massage tag	Massage tag description	Data description
0X1094/ 0X2094	0X00	BXP-device info packet upload	2 bytes, hex to binary Bit0: RSSI Bit1: timestamp Bit2: TX power Bit3: ranging data Bit4: ADV Interval Bit5: battery voltage Bit6: device property indicator Bit7: switch status indicator Bit8: firmware version Bit9: device name Bit10: advertising raw Bit11: response raw data

Read command:

App send: ef 20 94 dd 94 E2 66 56 11 00 00
Device reply: ef 20 94 dd 94 e2 66 56 11 00 05 00 00 02 03 ff

Description for the gateway reply:
Ef // head
20 94 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // BXP_device info upload tag
0002 // data length, 2 Byte
03ff // 0011 1111 1111 to indicate message in bit 0 to bit 9 are uploaded.

Set command:

App send: ef 10 94 dd 94 e2 66 56 11 00 05 00 00 02 03 ff
Device reply: ef 10 94 dd 94 e2 66 56 11 00 01 01 //command success

3.8.6 BXP-ACC packet(0X1095/0X2095)

To configure the uploading payload for the MOKO defined BXP-ACC packet.

Command flag	Massage tag	Massage tag description	Data description
0X1095/ 0X2095	0X00	BXP-ACC packet upload	2 bytes, hex to binary Bit0: RSSI Bit1: timestamp Bit2: TX power Bit3: ranging data Bit4: ADV interval Bit5: sampling data Bit6: full-scale Bit7: motion threshold Bit8: 3-axis data Bit9: battery voltage Bit10: advertising raw data

Read command:

App send: ef 20 95 dd 94 E2 66 56 11 00 00
Device reply: ef 20 95 dd 94 e2 66 56 11 00 05 00 00 02 03 ff

Description for the gateway reply:
Ef // head
20 95 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // BXP-ACC upload tag
0002 // data length, 2 Byte
03ff // 0011 1111 1111 to indicate message in bit 0 to bit 9 are uploaded.

Set command:

App send: ef 10 95 dd 94 e2 66 56 11 00 05 00 00 02 03 ff
Device reply: ef 10 95 dd 94 e2 66 56 11 00 01 01 //command success

3.8.7 BXP-TH packet(0X1096/0X2096)

To configure the uploading payload for the MOKO defined BXP-T&H packet.

Command flag	Massage tag	Massage tag description	Data description
0X1096/ 0X2096	0X00	BXP-TH packet upload	2 bytes, hex to binary Bit0: RSSI Bit1: timestamp

			Bit2: TX power Bit3: ranging data Bit4: ADV interval Bit5: temperature Bit6: humidity Bit7: battery voltage Bit8: advertising raw data

Read command:

App send: ef 20 96 dd 94 E2 66 56 11 00 00
Device reply: ef 20 96 dd 94 e2 66 56 11 00 05 00 00 02 00 ff

Description for the gateway reply:
Ef // head
20 96 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // BXP-TH upload tag
0002 // data length, 2 Byte
00ff // 0000 1111 1111 to indicate message in bit 0 to bit 7 are uploaded.

Set command:

App send: ef 10 96 dd 94 e2 66 56 11 00 05 00 00 02 00 ff
Device reply: ef 10 96 dd 94 e2 66 56 11 00 01 01 //command success

3.8.8 BXP-Button packet(0X1097/0X2097)

To configure the uploading payload for the MOKO defined BXP-Button packet.

Command flag	Massage tag	Massage tag description	Data description
0X1097/ 0X2097	0X00	BXP-button packet upload	4 bytes, hex to binary Bit0: RSSI Bit1: timestamp Bit2: frame type Bit3: status flag Bit4: Trigger count Bit5: Device id Bit6: Firmware type Bit7: Device name Bit8: Full-scale Bit9: motion threshold Bit10: 3-axis data Bit11: temperature Bit12: ranging data

			Bit13: battery voltage Bit14: TX power Bit15: advertising raw data Bit16: response raw data

Read command:

App send: ef 20 97 dd 94 E2 66 56 11 00 00
Device reply: ef 20 97 dd 94 e2 66 56 11 00 07 00 00 04 00 00 7f ff

Description for the gateway reply:
Ef // head
20 97 // command flag
dd 94e2 6656 11 // gateway MAC
00 07 // total data length, 7 Byte
00 // BXP_button upload tag
0004 // data length, 4 Byte
0000 7fff // 0111 1111 1111 1111 to indicate message in bit 0 to bit 14 are uploaded.

Set command:

App send: ef 10 97 dd 94 e2 66 56 11 00 07 00 00 04 00 00 7f ff
Device reply: ef 10 97 dd 94 e2 66 56 11 00 01 01 //command success

3.8.9 BXP-Tag packet(0X1098/0X2098)

To configure the uploading payload for the MOKO defined BXP-Tag packet.

Command flag	Massage tag	Massage tag description	Data description
0X1098/ 0X2098	0X00	BXP-Tag packet upload	2 bytes, hex to binary Bit0: RSSI Bit1: timestamp Bit2: sensor data Bit3: hall trigger event count Bit4: motion trigger event count Bit5: 3-axis data Bit6: battery voltage Bit7: tag id Bit8: device name Bit9: advertising raw data Bit10: response raw data

Read command:

App send: ef 20 98 dd 94 E2 66 56 11 00 00
Device reply: ef 20 98 dd 94 e2 66 56 11 00 05 00 00 02 01 ff

Description for the gateway reply:
Ef // head
20 98 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // BXP_tag upload tag
0002 // data length, 2 Byte
01ff // 0001 1111 1111 to indicate message in bit 0 to bit 8 are uploaded.

Set command:

App send: ef 10 98 dd 94 e2 66 56 11 00 05 00 00 02 01 ff
Device reply: ef 10 98 dd 94 e2 66 56 11 00 01 01 //command success

3.8.10 PIR packet(0X1099/0X2099)

To configure the uploading payload for the MOKO defined MK PIR packet.

Command flag	Massage tag	Massage tag description	Data description
0X1099/ 0X2099	0X00	PIR packet upload	2 bytes, hex to binary Bit0: RSSI Bit1: timestamp Bit2: PIR delay response status Bit3: door open/close status Bit4: sensor sensitivity Bit5: sensor detection status Bit6: battery voltage Bit7: major Bit8: minor Bit9: RSSI@1m Bit10: Tx power Bit11: adv name Bit12: advertising raw data Bit13: Response raw data

Read command:

App send: ef 20 99 dd 94 E2 66 56 11 00 00
Device reply: ef 20 99 dd 94 e2 66 56 11 00 05 00 00 02 0f ff

Description for the gateway reply:
Ef // head
20 99 // command flag
dd 94e2 6656 11 // gateway MAC
00 05 // total data length, 5 Byte
00 // PIR upload tag
0002 // data length, 1 Byte
0fff // 0000 1111 1111 1111 to indicate message in bit 0 to bit 11 are uploaded.

Set command:

App send: ef 10 99 dd 94 e2 66 56 11 00 05 00 00 02 0f ff
Device reply: ef 10 99 dd 94 e2 66 56 11 00 01 01 //command success

3.8.11 TOF packet(0X109A/0X209A)

To configure the uploading payload for the MOKO defined MK TOF packet.

Command flag	Massage tag	Massage tag description	Data description
0X109A/ 0X209A	0X00	TOF packet upload	1 byte, hex to binary Bit0: RSSI Bit1: timestamp Bit2: manufacturer vendor code Bit3: battery voltage Bit4: user data Bit5: ranging distance Bit6: advertising raw data

Read command:

App send: ef 20 9a dd 94 e2 66 56 11 00 00
Device reply: ef 20 9a dd 94 e2 66 56 11 00 04 00 00 01 3f

Description for the gateway reply:
Ef // head
20 9a // command flag
dd 94e2 6656 11 // gateway MAC
00 04 // total data length, 4 Byte
00 // TOF upload tag
0001 // data length, 1 Byte
3f // 0011 1111 to indicate message in bit 0 to bit 5 are uploaded.

Set command:

App send: ef 10 9a dd 94 e2 66 56 11 00 04 00 00 01 3f
Device reply: ef 10 9a dd 94 e2 66 56 11 00 01 01 //command success

3.8.12 Other type packet(0X109B/0X209B)

To configure the uploading payload for other type BLE advertising packet.

Command flag	Massage tag	Massage tag description	Data description
0X109B/ 0X209B	0X00	upload item selection	1 byte, hex to binary Bit0: RSSI Bit1: timestamp Bit2: advertising raw data Bit3: response raw data
	0X01	Upload data block. Each expression is 3 bytes, includes data type(1 byte), start byte(1 byte) and end byte(1 byte).	Data type: hex, 0x00~0xFF Start byte: number End byte: number The end byte should be no less than the start byte.

Read command:

App send: ef 20 9b dd 94 E2 66 56 11 00 00
Device reply: ef 20 9b dd 94 e2 66 56 11 00 0f 00 00 01 07 01 00 03 FF 01 05 01 00 03 16 01 09

Description for the gateway reply:
Ef // head
20 9b // command flag
Dd 94 e2 66 56 11 // gateway MAC
00 10 // total data length, 16 Byte
00 // upload item tag
00 01 // data length, 1 Byte
0f // 1111to indicate message in bit 0 to bit 3 are uploaded.
01 // data block tag
00 03 // data length, 3 byte
ff 01 05 // the first data block: data type: 0XFF, from 1 to 5 bytes will be uploaded.
01 // data block tag
00 03 // data length, 3 byte
16 01 09 // the second data block: data type: 0X16, from 1 to 9 bytes will be uploaded.

Set command:

App send: ef 10 9b dd 94 e2 66 56 11 00 04 00 00 01 03
Device reply: ef 10 9b dd 94 e2 66 56 11 00 01 01 //command success

3.9 BLE scan data reporting (0X30A0)

When the gateway BLE scan is enabled, it will actively report BLE scanning data to cloud. The gateway divides the BLE raw data into several types, uses type code to represent BLE data type in the payload:
0: iBeacon
1: Eddystone-UID
2: Eddystone-URL
3: Eddystone-TLM
4: BXP-Device info
5: BXP-ACC
6: BXP-TH
7: BXP-Button
8: BXP-Tag
9: PIR
10: Other type
11: TOF

The type 1-9 and type 11 are supported on MOKO Beacon. Except for these 10 types, all other BLE data will be identified as “other” type.

3.9.1 Report iBeacon packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 0: iBeacon
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	5 bytes, number, timestamp (4 bytes)+ time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	uuid	Hex, 16 bytes
	0X0B	major	number, 2 bytes
	0X0C	minor	number, 2 bytes

	0X0D	rssi@1m	number, 1 byte. value with sign

If the the gateway is set to separately upload iBeacon UUID, major, minor and RSSI @ 1m, the uploading payload will be as below:

```
Ef 30 a0 dd 94 e2 66 56 11 00 3e 00 00 01 00 01 00 06 f6 eb 5c 51 c4 1e 02 00 01 01 03 00 05 65 b8 ae 5b 00 04 00 01 ae 0a 00 10 11 22 33 44 55 66 77 88 99 00 11 22 33 44 55 66 0b 00 02 00 09 0c 00 02 00 09 0d 00 01 c5
```

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 3e // total data length, 62 Bytes
00 // type code tag
0001 // data length 1 Byte
00 // type code, iBeacon
01 // beacon MAC tag
0006 // data length 6 Byte
f6eb 5c51 c41e // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // Beacon is connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b8ae 5b00 // timestamp, 2024-01-30 16: 07: 55, time zone, 00
04 // RSSI tag
00 01 // data length 1 Byte
ae // RSSI, -82dBm
0a // uuid tag
00 10 // data length, 16 Byte
11 2233 4455 6677 8899 0011 2233 4455 66 // 16 bytes uuid
0b //major tag
0002 // data length, 2 Byte
0009 // major: 9
0c //minor tag
00 02 // data length, 2 Byte
00 09 // minor: 9
0d //RSSI@1m tag
0001 // data length, 1 Byte
c5 // RSSI@1m, -59dBm

If the the gateway is set to upload iBeacon raw advertising and response data, the uploading payload will be as below:

```
Ef 30 a0 dd 94 e2 66 56 11 00 5f 00 00 01 00 01 00 06 dc ba d1 9f dd 6f 02 00 01 01
03 00 05 65 b8 ae 5b 00 04 00 01 cc 05 00 1e 02 01 06 1a ff 4c 00 02 15 11 22 33 44
55 66 77 88 99 00 11 22 33 44 55 66 00 05 00 05 c5 06 00 1e 02 0a 00 1a 16 ab fe
50 c5 a 11 22 33 44 55 66 77 88 99 00 11 22 33 44 55 66 00 05 00 05
```

Description for the gateway reply:

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 5f // total data length, 95 Bytes
00 // type code tag
0001 // data length 1 Byte
00 // type code, iBeacon
01 // beacon MAC tag
0006 // data length 6 Byte
dc ba d1 9f dd 6f // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // Beacon is connectable
03 // timestamp tag
00 05 // data length 5 Bytes
65 b8 ae 5b 00 // timestamp, 2024-01-30 16: 07: 55, time zone, 00
04 // RSSI tag
00 01 // data length, 1 byte
cc // -52 dB
05 // advertising raw data tag
00 1e // data length 30 Bytes
02 01 06 1a ff 4c 00 02 15 11 22 33 44 55 66 77 88 99 00 11 22 33 44 55 66 00 05 00
05 c5 // 30 bytes iBeacon advertising raw data
06 //Response raw data tag
00 1e // data length, 30 Byte
02 0a 00 1a 16 ab fe 50 c5 0a 11 22 33 44 55 66 77 88 99 00 11 22 33 44 55 66 00 05
0005 // 30 bytes iBeacon response raw data

3.9.2 Report Eddystone-uid packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 1: eddystone-uid
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4

			bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	namespace	Hex, 10 bytes
	0X0B	rssi@0m	number, 1 byte. value with sign.
	0X0C	instance	Hex, 6 bytes

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 37 00 00 01 01 01 00 06 f4 b8 7c 6e 45 6d 02 00 01 01 03 00 05 65 b9 f8 65 00 04 00 01 b3 0b 00 01 00 0a 00 0a 11 22 33 44 45 55 55 66 66 67 0c 00 06 aa bb cc dd ee ff

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 37 // total data length, 55 Bytes
00 // type code tag
0001 // data length 1 Byte
01 // type code tag, Eddystone-uid
01 // beacon MAC tag
0006 // data length 6 Byte
f4b8 7c6e 456d // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9f8 6500 // 2024-01-31 15: 36: 05, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
b3 // RSSI -76dBm
0b // RSSI@0m tag
00 01 // data length 1 Byte
00 // RSSI@0m, 0
0a // namespace tag
00 0a // data length 10 Bytes
11 2233 4445 5555 6666 67 // 10 bytes namespace
0c // instance tag
0006 // data length 6 Bytes
aabb ccdd eeff // 6 bytes instance

3.9.3 Report Eddystone-url packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 2: eddystone-url
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte, half time zone)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	url	string
	0X0B	rssi@0m	number, 1 byte number with sign

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 36 00 00 01 02 01 00 06 cf fc 55 56 79 ca 02 00 01 01 03 00 05 65 b9 fa 45 00 04 00 01 ab 0a 00 14 68 74 74 70 3a 2f 2f 77 77 77 2e 52 6f 61 6d 62 65 65 0b 00 01 00

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 36 // total data length, 54 Bytes
00 // type code tag
0001 // data length 1 Byte
02 // type code, eddystone-url
01 // beacon MAC tag
0006 // data length 6 Byte
cffc 5556 79ca // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fa 4500 //2024-01-31 15: 44: 05, time zone 00
04 // RSSI tag
00 01// data length 1 Byte
ab // RSSI -84dBm

0a // url tag
00 14 // data length 20 Byte
68 74 74 70 3A 2F 2F 77 77 77 2E 6D 6F 6B 6F 73 6D 61 72 74 // url: http: //www.mokosmart
0b // rssi_0m tag
0001 // data length 1 Byte
0 // rssi_0m, 0

3.9.4 Report eddystone-tlm packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 3: eddystone-tlm
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte, half time zone)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	TLM version	number, 1 byte 0/1
	0X0B	Battery voltage	number, 2 bytes, unit:mV
	0X0C	temperature	number, 2 bytes Divide the value by 256 to obtain the actual temperature
	0X0D	ADV CNT	number, 4 bytes
	0X0E	SEC CNT	number, 4 bytes, unit 0.1 second

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 39 00 00 01 03 01 00 06 f4 05 04 03 02 02 02 00 01 00 03 00 05 65 b9 fa 8d 00 04 00 01 b8 0a 00 01 00 0b 00 02 0b 73 0c 00 02 15 fb 0d 00 04 01 45 4f 64 0e 00 04 01 56 8f 82

Description for the gateway reply:
Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC

00 39 // total data length, 57 Bytes
00 // type code tag
0001 // data length 1 Byte
03 // type code tag, eddystone-tlm
01 // beacon MAC tag
0006 // data length 6 Byte
f405 0403 0202 // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
00 // unconnectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fa 8d00 // 2024-01-31 15: 45: 17, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
b8 // RSSI -71dBm
0a // tlm version tag
00 01 // data length 1 Byte
00 // TLM version, 0
0b // battery voltage tag
00 02 // data length 2 Byte
0b 73 // battery voltage, 2931mV
0c // temperature tag
0002 // data length 2 Byte
15fb // 21℃
0d // ADV CNT tag
00 04 // data length 4 Byte
01 45 4f 64 // ADV CNT: 21319524
0e // SEC CNT tag
0004 // data length 4 Byte
0156 8f82 //SEC CNT: 2245005 seconds

3.9.5 Report BXP-device info packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 4: bxp-devinfo
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1

			byte)
	0X04	RSSI	number, 1 byte signed value
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	Tx power	number, 1 byte signed value
	0X0B	Ranging data	number, 1 byte. signed value, -100~0 dBm
	0X0C	ADV interval	number, 1 byte. unit 100ms
	0X0D	battery voltage	number, 2 bytes Unit mV
	0X0E	device property indicator	Hex, 1 byte Bit0-1: password verify status(00 enable ,10 disable) Bit2: ambient light sensor status Bit3: hall door sensor status
	0X0F	switch status indicator	Hex, 1 byte Bit0: connectable Bit1: ambient light status Bit2: door status
	0X10	firmware version	string
	0X11	device name	string

Notify command:

Device send: ef 30 a0 dd 94 e2 66 56 11 00 4a 00 00 01 04 01 00 06 e5 6b 1b 22 4a a6 02 00 01 01 03 00 05 65 b9 fa b8 00 04 00 01 ab 0a 00 01 00 0b 00 01 bf 0c 00 01 0a 0d 00 02 0b ca 0e 00 01 00 0f 00 01 01 10 00 06 56 32 2e 30 2e 31 11 00 08 47 57 54 45 53 54 31 31

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
004a // total data length, 74 Bytes
00 // type code tag
0001 // data length 1 Byte
04 // type code tag, bxp-device info

01 // beacon MAC tag
0006 // data length 6 Byte
e56b 1b22 4aa6 // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fa b800 // 2024-01-31 15: 46: 00, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
ab // RSSI, -84dBm
0a // tx power tag
00 01 // data length 1 Byte
00 // tx power: 0 dBm
0b // ranging data tag
00 01 // data length 1 Byte
bf // ranging data, -65dBm
0c // adv interval tag
00 01 // data length 1 Byte
0a // adv interval: 1 second
0d // battery voltage tag
00 02 // data length 2 Byte
0b ca // battery voltage: 3018mV
0e // device property indicator tag
0001 // data length 1 Byte
00 // password verify 0, ambient light sensor status 0, hall door sensor status 0
0f // switch status indicator tag
0001 // data length 1 Byte
01 // connectable, ambient light status on, door status on
10 // firmware version tag
0006 // data length 6 Byte
5632 2e30 2e31 //firmware version V2.0.1
11 // device name tag
00 08 // data length 8 Byte
47 5754 4553 5431//device name GWTEST11

3.9.6 Report BXP-ACC packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 5: BXP-ACC
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number,

			0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	TX power	number, 1 byte. Signed
	0X0B	ranging data	number, 1 byte. Signed
	0X0C	ADV interval	number, 1 byte. unit 100ms
	0X0D	sampling rate	number, 1 byte 0: 1hz 1: 10hz 2: 25hz 3: 50hz 4: 100hz
	0X0E	full-scale	number, 1 byte 0: 2g 1: 4g 2: 8g 3: 16g
	0X0F	motion threshold	number, 1 byte unit 0.1g
	0X10	3-axis data	number, 6 bytes 2 bytes X-axis + 2 bytes Y-axis + 2 bytes Z axis data, unit: mg
	0X11	battery voltage	number, 2 bytes unit: mV

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 43 00 00 01 05 01 00 06 ef 1f c1 05 45 dd 02 00 01 01 03 00 05 65 b9 fb 22 00 04 00 01 aa 0a 00 01 00 0b 00 01 c5 0c 00 01 0a 0d 00 01 01 0e 00 01 00 0f 00 01 01 10 00 06 ff c0 fe 40 2e 40 11 00 02 0b 5a

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
0043 // total data length, 67Bytes
00 // type code tag
0001 // data length 1 Byte

05 // type code tag, BXP-ACC
01 // beacon MAC tag
0006 // data length 6 Byte
ef1f c105 45dd // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fb 2200 //2024-01-31 15: 47: 46, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
aa // RSSI -85dBm
0a // tx power tag
00 01 // data length 1 Byte
00 // tx power: 0dBm
0b // ranging data tag
00 01 // data length 1 Byte
C5 // ranging data, -59dBm
0c // adv interval tag
00 01 // data length 1 Byte
0a // adv interval: 1 second
0d // sampling rate tag
00 01 // data length 1 Byte
01 // sample rate: 10hz
0e // full-scale tag
00 01 // data length 1 Byte
00 // full-scale: 2g
0f // motion threshold tag
00 01 // data length 1 Byte
01 // motion threshold: 0.1g
10 // 3-axis data tag
00 06 // data length 6 Byte
ff c0 fe 40 2e 40 //X-axis: 65,472mg, Y-axis: 65,088 mg, Z-axis: 11,840 mg
11 // battery voltage tag
0002 // data length 2 Byte
0b5a // 2906mV

3.9.7 Report BXP-TH packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 6: bxp-th
	0X01	MAC	Hex, 6 bytes

	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	TX power	number, 1 byte Signed
	0X0B	ranging data	number, 1 byte Signed
	0X0C	Adv interval	number, 1 byte unit 100ms
	0X0D	temperature	number, 2 bytes unit 0.1 °
	0X0E	humidity	number, 2 bytes unit 0.1%
	0X0F	battery voltage	number, 2 bytes Unit mV

Gateway report:
Ef 30 a0 dd 94 e2 66 56 11 00 38 00 00 01 06 01 00 06 c2 12 ec b7 e4 fd 02 00 01 01 03 00 05 65 b9 fb 6f 00 04 00 01 b0 0a 00 01 04 0b 00 01 00 0c 00 01 01 0d 00 02 00 e7 0e 00 02 03 e8 0f 00 02 0d 43

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
0038 // total data length, 56 Bytes
00 // type code tag
0001 // data length 1 Byte
06 // type code tag, bxp-th
01 // beacon MAC tag
0006 // data length 6 Byte
c212 ecb7 e4fd // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte

65 b9fb 6f00 // 2024-01-31 15: 49: 03, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
b0 // RSSI -80dBm
0a // tx power tag
00 01 // data length 1 Byte
04 // tx power: 4dBm
0b // ranging data tag
00 01 // data length 1 Byte
00 // ranging data: 0 dBm
0c // adv interval tag
00 01 // data length 1 Byte
01 // adv interval:100ms
0d // temperature tag
00 02 // data length 2 Byte
00 e7 //temperature: 23.1℃
0e // humidity tag
0002 // data length 2 Byte
03e8 // humidity: 100%
0f // battery voltage tag
00 02 // data length 2 Byte
0d 43 // battery voltage: 3395mV

3.9.8 Report BXP-Button packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 7: bxp-button
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	frame type	hex, 1 byte 0X20: single press mode 0X21: double press

			mode 0X22: long press mode 0X23: abnormal inactivity mode
	0X0B	status flag	Hex, 1 byte Bit0: password vertify flag Bit1: trigger status
	0X0C	trigger count	number, 2 bytes
	0X0D	device id	Hex, 1-6 bytes
	0X0E	firmware type	Hex, 1 byte
	0X0F	device name	string
	0X10	full scale	number, 1 byte 0: 2g 1: 4g 2: 8g 3: 16g
	0X11	motion threshold	number, 2 bytes Unit 0.1g
	0x12	3-axis data	number, 6 bytes 2 bytes X-axis + 2 bytes Y-axis + 2 bytes Z axis data, unit: mg
	0x13	temperature	number, 2 bytes Unit 0.25 °
	0x14	ranging data	number, 1 byte Signed value
	0x15	battery voltage	number, 2 bytes Unit:mV
	0x16	TX power	Hex, 1 byte Signed value

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 42 00 00 01 07 01 00 06 dc 2d da 6e 2b 4e 02 00 01 01 03 00 05 65 b9 fb 98 00 04 00 01 b5 0a 00 01 20 0b 00 01 fa 0c 00 02 0b 90 0d 00 05 0b 7f 00 f4 00 0e 00 01 5b 0f 00 09 4d 4b 2042 75 74 74 6f 6e

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 42 // total data length, 66 Bytes
00 // type code tag
0001 // data length 1 Byte
07 // type code tag, bxp-button
01 // beacon MAC tag
0006 // data length 6 Byte
dc2d da6e 2b4e // beacon MAC

02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fb 9800 //2024-01-31 15: 49: 44, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
b5 // RSSI, -74dBm
0a // frame type tag
00 01 // data length 1 Byte
20 // single press mode
0b // status flag tag
00 01 // data length 1 Byte
fa // password verify flag 0, trigger status 1
0c // trigger count tag
00 02 // data length 2 Byte
0b 90 // trigger count: 2960
0d // device id tag
0005 // data length 5 Byte
0b7f 00f4 00 // device id
0e // firmware type tag
0001 // data length 1 Byte
5b // firmware type
0f // device name tag
0009 // data length 9 Byte
4d4b 2042 7574 746f 6e // MK Button

3.9.9 Report BXP-Tag packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 8: bxp-tag
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined

	0X06	response raw data	Hex. length is not defined
	0X0A	sensor status	hex, 1 byte Bit0: hall sensor status bit1: accelerometer sensor status Bit2: accelerometer sensor equipped status
	0X0B	hall trigger event count	number, 2 bytes
	0X0C	motion trigger event count	number, 2 bytes
	0X0D	3-axis data	number, 6 bytes 2 bytes X-axis + 2 bytes Y-axis + 2 bytes Z axis data, unit: mg
	0X0E	battery voltage	number, 2 bytes Unit mV
	0X0F	tagid	Hex, 1-6 bytes
	0X10	device name	string

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 3f 00 00 01 08 01 00 06 1c 34 f1 55 5f 64 02 00 01 01 03 00 05 65 b9 fb db 00 04 00 01 b4 0a 00 01 07 0b 00 02 00 10 0c 00 02 00 0a 0d 00 06 ff c0 fe 40 2e 40 0e 00 02 0b 5a 0f 00 03 00 00 01

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 3f // total data length, 63 Byte
00 // type code tag
0001 // data length 1 Byte
08 // type code tag, bxp-tag
01 // beacon MAC tag
0006 // data length 6 Byte
1c34 f155 5f64 // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fb db00 //2024-01-31 15: 50: 51, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
b4 // RSSI -75dBm
0a // Sensor status tag

00 01 // data length 1 Byte
07 //hall sensor status 1, accelerometer sensor status 1, accelerometer sensor equipped status 1
0b // hall trigger event count tag
00 02 // data length 2 Byte
00 10 //hall trigger event count: 16
0c //motion trigger event count tag
0002 // data length 1 Byte
000A //motion trigger event count: 10
0d // 3-axis data tag
00 06 // data length 6 Byte
ff c0 fe 40 2e 40 //X-axis: 65,472mg, Y-axis: 65,088 mg, Z-axis: 11,840 mg
0e // battery voltage tag
0002 // data length 2 Byte
0b5a // 2906mV
0f // tag id tag
00 03 // data length 3 Byte
00 0001 //tag id: 000001

3.9.10 Report MK PIR packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Code type	1 byte, number 9: PIR
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte, half time zone)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	delay response status	number, 1 byte 0: low delay 1: medium delay 2: high delay
	0X0B	door open/close status	number, 1 byte 0: close 1: open
	0X0C	sensor sensitivity	number, 1 byte

			0: low 1: medium 2: high
	0X0D	sensor detection status	number, 1 byte 0: no effective motion 1: effective motion
	0X0E	battery voltage	number, 2 bytes unit mV
	0X0F	major	number, 2 bytes
	0X10	minor	number, 2 bytes
	0X11	rssi@1m	number, 1 byte Signed value
	0x12	Tx power	number, 1 byte Signed value
	0x13	Adv name	string

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 40 00 00 01 09 01 00 06 ea 63 46 a0 f4 c1 02 00 01 01 03 00 05 65 b9 fc 12 00 04 00 01 ae 0a 00 01 00 0b 00 01 01 0c 00 01 00 0d 00 01 01 0e 00 02 0e 0a 0f 00 02 00 00 10 00 02 00 00 11 00 01 b0

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 40 // total data length, 64 Byte
00 // type code tag
0001 // data length 1 Byte
09 // type code tag, pir
01 // beacon MAC tag
0006 // data length 6 Byte
ea63 46a0 f4c1 // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fc 1200 //2024-01-31 15: 51: 46, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
ae // RSSI -82dBm
0a // PIR delay response status tag
00 01 // data length 1 Byte
00 // low delay
0b // door open/close status tag
00 01 // data length 1 Byte
01 // Door open

0c // sensor sensitivity tag
00 01 // data length 1 Byte
00 // low sensitivity
0d // sensor detection status tag
00 01 // data length 1 Byte
01 // effective motion detected
0e // battery voltage tag
00 02 // data length 2 Byte
0e 0a // 3594mV
0f // major tag
0002 // data length 2 Byte
0000 // major:0
10 // minor tag
00 02 // data length 2 Byte
00 00 // minor: 0
11 // rssi@1m tag
0001 // data length 1 Byte
b0 // rssi@1m, -80dBm

3.9.11 Report Other type packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 10: other type
	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte, half time zone)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	1st data block data	Hex
	0X0B	2nd data block data	Hex
	0X0C	3rd data block data	Hex
	0X0D	4th data block data	Hex
	0X0E	5th data block data	Hex
	0X0F	6th data block data	Hex
	0X10	7th data block data	Hex
	0X11	8th data block data	Hex

	0X12	9th data block data	Hex
	0X13	10th data block data	Hex

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 31 00 00 01 0b 01 00 06 fa 37 c2 03 2f 92 02 00 01 01 03 00 05 65 b9 fe 2b 00 04 00 01 ab 0a 00 02 00 59 0b 00 02 0b bf 0c 00 02 ff ff 0d 00 02 02 16

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 31 // total data length, 49 Byte
00 // type code tag
0001 // data length 1 Byte
0b // type code tag, TOF
01 // beacon MAC tag
0006 // data length 6 Byte
fa37 c203 2f92 // beacon MAC
02 // connectable tag
00 01 // data length 1 Byte
01 // connectable
03 // timestamp tag
00 05 // data length 5 Byte
65 b9fe 2b00 //2024-01-31 16: 00: 43, time zone 00
04 // RSSI tag
00 01 // data length 1 Byte
ab //RSSI, -85dBm
0a // The 1st data block data tag
00 02 // data length 2 Byte
00 59 // The 1st data block data
0b // The 2nd data block data tag
0002 // data length 2 Byte
0bbf // The 2nd data block data
0c // The 3rd data block data tag
00 02 // data length 2 Byte
ff ff // The 3rd data block data
0d // The 4th data block data tag
0002 // data length 2 Byte
0216 // The 4th data block data

3.9.12 Report MK TOF packet

Command flag	Massage tag	Massage tag description	Data description
0X30A0	0X00	Type code	1 byte, number 10: other

	0X01	MAC	Hex, 6 bytes
	0X02	connectable	1 byte, number, 0: not connectable 1: connectable
	0X03	timestamp	number, 5 bytes Timestamp (4 bytes)+time zone (1 byte)
	0X04	RSSI	number, 1 byte
	0X05	advertising raw data	Hex. length is not defined
	0X06	response raw data	Hex. length is not defined
	0X0A	MFG code	Hex, 2 bytes
	0X0B	battery voltage	number, 2 bytes unit mV
	0X0C	user data	Hex, 2 bytes
	0X0D	ranging distance	number, 2 bytes unit: mm

Gateway report: ef 30 a0 dd 94 e2 66 56 11 00 30 00 00 01 0a 01 00 06 71 b5 31 8e 7f 26 02 00 01 00 03 00 05 65 b9 fd c8 00 04 00 01 b0 0a 00 02 59 00 0b 00 02 0b b8 0c 00 02 ff ff 0d 00 02 3c 00

Ef // head
30 a0 // command flag
dd 94e2 6656 11 // gateway MAC
00 30 // total data length, 48 Byte
00 // type code tag
01 // data length 1 Byte
0a // type code tag, other
00 // beacon MAC tag
0006 // data length 6 Byte
71b5 318e 7f26 // beacon MAC
02 //connectable tag
00 01 // data length 1 Byte
00 // un-connectable
03 //timestamp tag
00 05 // data length 5 Bytes
65 b9fd c800 //2024-01-31 15: 59: 04, time zone 00
04 //RSSI tag
00 01 // data length 1 Byte
b0 // RSSI -79dBm
0a // MFG code tag
00 02 // data length 2 bytes

59 00 // MFG code
0b // battery voltage flag
00 02 // data length 2 bytes
0b b8 //battery voltage: 3000mV
0c // user data tag
00 02 // data length 2 bytes
ff ff // user data
0d // ranging distance tag
00 02 // data length 2 bytes
3c 00 // ranging distance: 15,360 mm

3.10 Buffer Data uploading and operation

3.10.1 Buffer data operation (0X10B0)

Command flag	Massage tag	Massage tag description	Data description
0X10B0	0X00	operation	number 1 byte 0: start buffer data uploading 1: stop buffer data uploading 2: restore offline data uploading 3: clear buffer data

Set command:

App send: ef 10 B0 dd 94 E2 66 56 11 00 04 00 00 01 00 // offline data upload begins
Device reply: ef 10 B0 dd 94 E2 66 56 11 00 01 01 //command success

3.10.2 Buffer GPS data reporting(0X30B1)

The command flag is 0X30B1, the data structure is the same as chapter *3.6.10 GNSS data reporting.*

3.10.3 Buffer BLE data reporting (0X30B2)

The command flag is 0X30B2, the data structure is the same as chapter *3.9 BLE scan data reporting.*

3.10.4 Buffer data upload completed (0X30B3)

When the buffer data finishes to be uploaded, the gateway will report this message.

Command flag	Massage tag	Massage tag description	Data description
0X30B3	0X00	time stamp	number, 4 bytes

	0X01	uploaded packets number	number, 2 bytes
	0X02	remaining packets number	number, 2 bytes

Gateway report: ef 30 B3 dd 94 e2 66 56 11 00 0f 00 00 04 65 BF 34 5A 01 00 01 64 02 00 01 00

Ef // head
30 B3 // command flag
dd 94e2 6656 11 // gateway MAC
00 0f // total data length, 15 Bytes
00 // time stamp tag
0004 // data length, 4 Bytes
65BF 345A // 2024-02-04 14:53:14
01 // uploaded packet number flag
00 01 // data length, 1 Byte
64 // uploaded packet number: 100
02 // remaining packet number tag
00 01 // data length, 1 Byte
00 // remaining packet number: 0

3.11 Connect and control MOKO Button device

3.11.1 Connect with MOKO Button

To make the gateway connect with a certain MOKO Button device.

Command flag	Massage tag	Massage tag description	Data description
0X1100	0X00	BLE MAC	hex, 6 bytes
	0X01	BLE password	string, 0-16 characters

Send to gateway : ef 11 00 d9 6a 36 75 e9 f3 00 14 00 00 06 cb f6 e4 1a 4f e4 01 00 08 31 32 33 34 35 36 37 38
Gateway reply: ef 11 00 d9 6a 36 75 e9 f3 00 01 01

Ef // head
11 00 // command flag
d9 6a 36 75 e9 f3 // gateway MAC
00 14 // total data length, 20 Bytes
00 // BLE mac tag
00 06 // data length, 6 Bytes
cb f6 e4 1a 4f e4 // BLE mac
01 // BLE password tag
00 08 // data length, 8 Bytes

31 32 33 34 35 36 37 38// BLE password: 12345678

3.11.2 Notify connection result

After the gateway connects to the BLE Button device, it will notify the connection result to cloud.

<table>
<tr><th>Command flag</th><th>Massage tag</th><th>Massage tag description</th><th>Data description</th></tr>
<tr><td rowspan="13">0X3101</td><td>0X00</td><td>BLE MAC</td><td>hex, 6 bytes</td></tr>
<tr><td>0X01</td><td>result code</td><td>number.
0: Connect succeed
1: Connection is not allowed on current mode
2:Exceed the connection number
3:BLE device has been connected
4: BLE device is out of range
5: BLE device is not connectable
6: Connect failed
7: Device type error
8: Password error</td></tr>
<tr><td>0X02</td><td>product model</td><td>string</td></tr>
<tr><td>0X03</td><td>company name</td><td>string</td></tr>
<tr><td>0X04</td><td>hardware version</td><td>string</td></tr>
<tr><td>0X05</td><td>software version</td><td>string</td></tr>
<tr><td>0X06</td><td>firmware version</td><td>string</td></tr>
<tr><td>0X07</td><td>battery voltage</td><td>number, 2 bytes
unit: mV</td></tr>
<tr><td>0X08</td><td>single alarm number</td><td>number, 2 bytes</td></tr>
<tr><td>0X09</td><td>double alarm number</td><td>number, 2 bytes</td></tr>
<tr><td>0X0A</td><td>long press alarm number</td><td>number, 2 bytes</td></tr>
<tr><td>0X0B</td><td>alarm status</td><td>1 byte.
bit 0: single press alarm
bit 1: double press alarm
bit 2: long press alarm
bit 3: abnormal inactivity alarm</td></tr>
</table>

Gateway report: ef 31 01 d9 6a 36 75 e9 f3 00 66 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00 02 00 06 42 55 54 54 4f 4e 03 00 14 4d 4f 4b 4f 20 54 45 43 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e 04 00 0b 4d 4b 42 4e 20 53 65 72 69 65 73 05 00 07 42 58 50 2d 42 2d 44 06 00 06 56 31 2e 30 2e 38 07 00 02 0b ee 08 00 02 00 64 09 00 02 00 00 0a 00 02 00 00 0b 00 01 00

Ef // head
31 01// command flag
d9 6a 36 75 e9 f3 // gateway mac
00 66 // total data length, 102 bytes
00 //BLE mac tag
00 06 // data length, 6 bytes
cb f6 e4 1a 4f e4 // BLE MAC
01 //result code
00 01 //data length, 1 byte
00 //connect succeed
02 // product model tag
00 06 // data length, 6 bytes
42 55 54 54 4f 4e //Product model: BUTTON
03 // company name tag
00 14 // data length, 20 bytes
4d 4f 4b 4f 20 54 45 43 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e // company name: MOKO TECHNOLOGY LTD.
04 // Hardware version tag
00 0b // data length: 11 bytes
4d 4b 42 4e 20 53 65 72 69 65 73 //Hardware version: MKBN Series
05 //Software version tag
00 07 //data length: 7 bytes
42 58 50 2d 42 2d 44 //Software version: BXP-B-D
06 //Firmware version tag
00 06 // data length: 6 bytes
56 31 2e 30 2e 38 // firmware version: V1.0.8
07 // battery voltage tag
00 02 // 2 bytes
0b ee // battery voltage: 3054 mV
08 // single press alarm number tag
00 02 // data length
00 64 // single press alarm number: 100
09 //double press alarm number tag
00 02 //data length
00 00 //double press alarm number: 0
0a // long press alarm number tag
00 02 //data length
00 00 //long press alarm number: 0
0b // alarm status tag
00 01 // data length
00 // not triggered

3.11.3 Obtain BLE device information

To inquire device information from the BLE Button device.

Command flag	Massage tag	Massage tag description	Data description
0X1102	0X00	BLE MAC	hex, 6 bytes

Send to gateway : ef 11 02 d9 6a 36 75 e9 f3 00 09 00 00 06 cb f6 e4 1a 4f e4
Gateway reply: ef 11 02 d9 6a 36 75 e9 f3 00 01 01 // command success

3.11.4 Notify BLE device information

When the gateway obtains the device information from the Button device, it will transfer the information to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3103	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	number. 0: success 1: BLE device is not connected 2: Command is not allowed in current mode
	0X02	product model	string
	0X03	company name	string
	0X04	hardware version	string
	0X05	software version	string
	0X06	firmware version	string

Gateway report: ef 31 03 d9 6a 36 75 e9 f3 00 4E 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00 02 00 06 42 55 54 54 4f 4e 03 00 14 4d 4f 4b 4f 20 54 4 543 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e 04 00 0b 4d 4b 42 4e 20 53 65 72 69 65 73 05 00 07 42 58 50 2d 42 2d 44 06 00 06 56 31 2e 30 2e 38

Ef // head
31 03// command flag
d9 6a 36 75 e9 f3 // gateway mac
00 4e // total data length, 78 bytes
00 //BLE mac tag
00 06 // data length, 6 bytes
cb f6 e4 1a 4f e4 // BLE MAC
01 //result code
00 01 //data length, 1 byte
00 //connect succeed
02 // product model tag
00 06 // data length, 6 bytes

42 55 54 54 4f 4e //Product model: BUTTON
03 // company name tag
00 14 // data length, 20 bytes
4d 4f 4b 4f 20 54 45 43 48 4e 4f 4c 4f 47 59 20 4c 54 44 2e // company name: MOKO TECHNOLOGY LTD.
04 // Hardware version tag
00 0b // data length: 11 bytes
4d 4b 42 4e 20 53 65 72 69 65 73 //Hardware version: MKBN Series
05 //Software version tag
00 07 //data length: 7 bytes
42 58 50 2d 42 2d 44 //Software version: BXP-B-D
06 //Firmware version tag
00 06 // data length: 6 bytes
56 31 2e 30 2e 38 // firmware version: V1.0.8

3.11.5 Obtain BLE device status

To inquire device status from the BLE Button device.

Command flag	Massage tag	Massage tag description	Data description
0X1104	0X00	BLE MAC	hex, 6 bytes

Send to gateway : ef 11 04 d9 6a 36 75 e9 f3 00 09 00 00 06 cb f6 e4 1a 4f e4
Gateway reply: ef 11 04 d9 6a 36 75 e9 f3 00 01 01 // command success

3.11.6 Notify BLE device status

When the gateway obtains the device status information from the Button device, it will transfer the information to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3105	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	number. 0: Command success 1: BLE device is not connected 2: Command is not allowed in current mode
	0X02	battery voltage	number, 2 bytes unit: mV
	0X03	single alarm number	number, 2 bytes
	0X04	double alarm number	number, 2 bytes
	0X05	long press alarm number	number, 2 bytes
	0X06	alarm status	1 byte. bit 0: single press alarm

			bit 1: double press alarm bit 2: long press alarm bit 3: abnormal inactivity alarm

Gateway report: ef 31 05 d9 6a 36 75 e9 f3 00 25 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00 02 00 02 0b ee 03 00 02 00 64 04 00 02 00 0a 05 00 02 00 00 06 00 01 03

Ef // head
31 05 //command flag
d9 6a 36 75 e9 f3 // gateway MAC
00 25 // total data length, 37 bytes
00 // BLE MAC tag
00 06 // data length, 6 bytes
cb f6 e4 1a 4f e4 // BLE Mac
01 //result code tag
00 01 // data length: 1 byte
00 // command success
02 // battery voltage tag
00 02 // data length
0b ee //battery voltage: 3054 mV
03 //single press alarm tag
00 02 // data length
00 64 // single press alarm number: 100
04 //double press alarm number tag
00 02 //data length
00 0a //double press alarm number: 10
05 // long press alarm number tag
00 02 //data length
00 00 //long press alarm number: 0
06 //alarm status tag
00 01 // data length
03 //0011 to indicate single press alarm is triggered, double press alarm is triggered, long press alarm is not triggered, abnormal inactivity alarm is not triggered

3.11.7 Dismiss alarm status

To make the gateway control the Button dismiss alarm status.

Command flag	Massage tag	Massage tag description	Data description
0X1106	0X00	BLE MAC	hex, 6 bytes

Send to gateway : ef 11 06 d9 6a 36 75 e9 f3 00 09 00 00 06 cb f6 e4 1a 4f e4
Gateway reply: ef 11 06 d9 6a 36 75 e9 f3 00 01 01 // command success

3.11.8 Notify dismiss alarm result

After the gateway get reply from Button for the dismissing alarm status command, the gateway will transfer the reply to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3107	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	number. 0: Command success 1: BLE device is not connected 2: Command is not allowed in current mode

Gateway report: ef 31 07 d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00

3.11.9 Notify BLE disconnection

When the gateway is disconnected from the BLE Button, it will report a message to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3108	0X00	BLE MAC	hex, 6 bytes
	0X01	reason for disconnection	number 0: Abnormally disconnect 1: Communicate timeout

Gateway report: ef 31 08 d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00

3.11.10 Control Button LED

To make the gateway control Button LED.

Command flag	Massage tag	Massage tag description	Data description
0X1109	0X00	BLE MAC	hex, 6 bytes
	0X01	LED flash duration	number. 2 bytes range: 1-6000, unit: 0.1s
	0X02	LED flash interval	number. 2 bytes range: 0-10000, unit: 0.1s

Send to gateway : ef 11 09 d9 6a 36 75 e9 f3 00 13 00 00 06 cb f6 e4 1a 4f e4 01 00 02 00 0a 02 00 02 00 64
Gateway reply: ef 11 09 d9 6a 36 75 e9 f3 00 01 01 // command success

Description for the setup command:
ef // head
11 09 //command flag
d9 6a 36 75 e9 f3 // gateway MAC
00 13 // total data length: 19 bytes
00 //BLE mac tag
00 06 // data length: 6 bytes
cb f6 e4 1a 4f e4 // BLE MAC
01 // LED flash duration tag
00 02 // data length, 2 bytes
00 0a // LED flash duration: 1 seconds
02 // LED flash interval tag
00 02 //data length, 2 bytes
00 64 //LED flash duration: 10 seconds

3.11.11 Notify LED controlling result

After the gateway get reply from Button for the LED controlling command, the gateway will transfer the reply to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X310A	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	number. 0: Command success 1: BLE device is not connected 2: Command is not allowed in current mode

Gateway report: ef 31 0A d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00

3.11.12 Control Button Buzzer

To make the gateway control Button Buzzer.

Command flag	Massage tag	Massage tag description	Data description
0X110B	0X00	BLE MAC	hex, 6 bytes
	0X01	Buzzer ring duration	number. 2 bytes range: 1-6000, unit: 0.1s
	0X02	Buzzer ring interval	number. 2 bytes range: 0-10000, unit: 0.1s

Send to gateway : ef 11 0b d9 6a 36 75 e9 f3 00 13 00 00 06 cb f6 e4 1a 4f e4 01 00 02 00 64 02 00 02 00 0a
Gateway reply: ef 11 0b d9 6a 36 75 e9 f3 00 01 01 // command success

Description for the setup command:
ef // head
11 0b //command flag
d9 6a 36 75 e9 f3 // gateway MAC
00 13 // total data length: 19 bytes
00 //BLE mac tag
00 06 // data length: 6 bytes
cb f6 e4 1a 4f e4 // BLE MAC
01 // Buzzer ring duration tag
00 02 // data length, 2 bytes
00 0a // Buzzer ring duration: 1 seconds
02 // Buzzer ring interval tag
00 02 //data length, 2 bytes
00 64 //Buzzer ring interval: 10 seconds

3.11.13 Notify Buzzer controlling result

After the gateway get reply from Button for the Buzzer controlling command, the gateway will transfer the reply to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X310C	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	number. 0: Command success 1: BLE device is not connected 2: Command is not allowed in current mode

Gateway report: ef 31 0c d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00

3.11.14 Disconnect from Button device

To make the gateway disconnect from the MOKO Button device.

Command flag	Massage tag	Massage tag description	Data description
0X1150	0X00	BLE MAC	hex, 6 bytes

Send to gateway : ef 11 50 d9 6a 36 75 e9 f3 00 09 00 00 06 cb f6 e4 1a 4f e4
Gateway reply: ef 11 50 d9 6a 36 75 e9 f3 00 01 01 // command success

3.11.15 Upgrade BLE Button firmware

To upgrade the Button firmware via gateway.

Command flag	Massage tag	Massage tag description	Data description
0X1152	0X00	BLE MAC	hex, 6 bytes
	0X01	firmware URL	string. 1-256 characters
	0X02	initial data URL	string. 1-256 characters

Send to gateway:
ef 11 52 dd 94 e2 66 56 11 00 84 00 00 06 cb f6 e4 1a 4f e4 01 00 37 68 74 74 70 3a 2f 2f 34 37 2e 31 30 34 2e 31 37 32 2e 31 36 39 3a 38 30 38 30 2f 75 70 64 61 74 61 5f 66 6f 6c 64 2f 42 58 50 5f 42 5f 56 31 2e 30 2e 32 2e 62 69 6e 02 00 3e 68 74 74 70 3a 2f 2f 34 37 2e 31 30 34 2e 31 37 32 2e 31 36 39 3a 38 30 38 30 2f 75 70 64 61 74 61 5f 66 6f 6c 64 2f 69 6e 69 74 69 61 6c 5f 64 61 74 61 5f 56 31 2e 30 2e 32 2e 62 69 6e

Description for the setup command:
ef // head
11 52 // command flag
dd 94 e2 66 56 11 // gateway MAC
00 84 // Total data length, 132 bytes
00 // BLE MAC tag
00 06 // data length
cb f6 e4 1a 4f e4 //BLE MAC
01 //firmware URL flag
00 37 // firmware URL length, 55 bytes
68 74 74 70 3a 2f 2f 34 37 2e 31 30 34 2e 31 37 32 2e 31 36 39 3a 38 30 38 30 2f 75 70 64 61 74 61 5f 66 6f 6c 64 2f 42 58 50 5f 42 5f 56 31 2e 30 2e 32 2e 62 69 6e
// firmware URL: http://47.104.172.169:8080/updata_fold/BXP_B_V1.0.2.bin
02 // initial URL flag
00 3e // initial data URL length, 62 bytes
68 74 74 70 3a 2f 2f 34 37 2e 31 30 34 2e 31 37 32 2e 31 36 39 3a 38 30 38 30 2f 75 70 64 61 74 61 5f 66 6f 6c 64 2f 69 6e 69 74 69 61 6c 5f 64 61 74 61 5f 56 31 2e 30 2e 32 2e 62 69 6e
// initial data URL: http://47.104.172.169:8080/updata_fold/initial_data_V1.0.2.bin

3.11.16 Notify BLE DFU process

When the Button DFU starts, the gateway will obtain the upgrade process percentage every second and report it to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3153	0X00	BLE MAC	hex, 6 bytes
	0X01	percent	number. unit: %

Gateway report: ef 31 53 d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 0a

3.11.17 Notify BLE DFU result

When the Button DFU finishes, the gateway will obtain the upgrade result and report it to cloud.

Command flag	Massage tag	Massage tag description	Data description
0X3154	0X00	BLE MAC	hex, 6 bytes
	0X01	result code	0: success 1: BLE button is not connected 2: command is not allowed in current mode 3: failed to obtain upgrade files 4: command timeout 5: upgrade failed

Gateway report: ef 31 54 d9 6a 36 75 e9 f3 00 0d 00 00 06 cb f6 e4 1a 4f e4 01 00 01 00

4. Revision History

Revision	Description	Editor	Date
V1.0	Initial version	weiguifen	2024.2.5
V1.1	Improved document, added some chapters: *3.5.6 Raw data filtration* *3.8 Upload payload settings* *3.10 Buffer data uploading and operation*	weiguifen	2024.3.9
V1.2	Add chapter *3.11 Connect and control MOKO Button device*	weiguifen	2024.4.18

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